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# Evaluation of the psychometric properties of patient- and clinician-reported outcome measures of chemotherapy-induced peripheral neuropathy: a COSMIN systematic review protocol



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| Complete List of Authors: | Bérubé-Mercier, Philippe; Université Laval, Faculty of Nursing; CHU de Quebec-Universite Laval, Oncology Division<br>Tapp, Diane; Laval University, Faculty of Nursing; Quebec Heart and Lung Institute<br>Cimon, Marie-Ève; Université Laval, Faculty of Nursing; CHU de Quebec-Universite Laval, Oncology Division<br>Li, Tiffany; The University of Sydney, School of Medical Sciences, Brain and Mind Center<br>Park, Susanna; The University of Sydney, School of Medical Sciences, Brain and Mind Centre<br>Bouhêlier, Éve; CHU de Quebec-Universite Laval<br>McGarragle, Kaitlin; CHU de Quebec-Universite Laval, Oncology Division<br>Robichaud, Lye-Ann; Université de Montréal, Department of Psychology<br>Gewandter, Jennifer; University of Rochester Medical Center, School of Medicine and Dentistry<br>Bouchard, Maxime<br>Gauthier, Lynn; Université Laval, CHU de Quebec-Université Laval Research Center, Oncology Division; Université Laval, Department of Family and Emergency Medicine |
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Title: Evaluation of the psychometric properties of patient- and clinician-reported outcome measures of chemotherapy-induced peripheral neuropathy: a COSMIN systematic review protocol

Authors:

Philippe Bérubé-Mercier [1,2,3], Diane Tapp[1,2,4], Marie-Ève Cimon [1,2,3,4,] Tiffany Li [5], Susanna B. Park [5], Ève Bouhêlier [3], Kaitlin McGarragle [3], Lye-Ann Robichaud [6], Jennifer Gewandter[7], Maxime Bouchard, Lynn R. Gauthier [2,3,8]

1. Faculty of Nursing, Université Laval, Quebec City, Quebec, Canada
2. Équipe de recherche Michel-Sarrazin en oncologie psychosociale et soins palliatifs (ERMOS), Quebec City, Quebec, Canada.
3. CHU de Québec-Université Laval Research Center, Oncology Division, Quebec City, Quebec, Canada
4. Centre de recherche de l'Institut Universitaire de Cardiologie et de Pneumologie de Québec-Université Laval, Quebec City, Quebec, Canada
5. Faculty of Medicine and Health, School of Medical Sciences, Brain and Mind Centre, The University of Sydney, Sydney, Australia
6. Department of Psychology, Faculty of Arts and Science, Université de Montréal, Montreal, Canada.
7. School of Medicine and Dentistry, University of Rochester Medical Center, Rochester, New York, United States
8. Department of Family and Emergency Medicine, Faculty of Medicine, Université Laval, Quebec City, Quebec, Canada

Contact Author : Lynn R. Gauthier

Hôtel-Dieu de Québec - CRCEO
6, rue McMahon, local 1899-5
Quebec City, Québec
Canada G1R 3S1
Email : Lynn.Gauthier@crchudequebec.ulaval.ca

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## Abstract

**Introduction:** Chemotherapy-induced peripheral neuropathy (CIPN) is a poorly understood side effect of many antineoplastic agents. Patients may experience sensory, motor, and autonomic symptoms, which negatively impact quality of life. A gold-standard assessment methodology has yet to be determined, limiting efforts to identify effective agents to prevent or treat CIPN.

**Methods and analysis:** This is a protocol of a systematic review of psychometric analyses of CIPN Clinician Reported Outcome Measures (ClinROM) and Patient-Reported Outcome Measures (PROM) among adults receiving, or who had previously received chemotherapy for cancer. The COnsensus-based Standards for the selection of health Measurement INstruments (COSMIN) quality ratings will be compared across studies and across ClinROMs and PROMs. Studies reporting psychometric proprieties of CIPN ClinROMs and/or PROMs among adults aged ≥ 18 years will be eligible for inclusion, with no restriction on language or year of publication. MEDLINE, Embase, CINAHL, and APA PsycInfo databases will be searched. Study characteristics, measurement properties of the ClinROMs and/or PROMs and the CIPN definitions will be extracted. The Synthesis Without Meta-analysis guideline will be used to guide data synthesis. The COSMIN Risk of Bias checklist will be used by two independent raters to assess methodological quality. A subgroup analyses by age, chemotherapy type, and the timing of the study in relation to the delivery of chemotherapy will be carried out where data are available. An adapted version of Outcome Measures in Rheumatology (OMERACT) filter 2.1 will be used to provide a best-evidence synthesis of CIPN ClinROMs and PROMs and to recommend a CIPN assessment tool for clinical and research settings.

**Ethics and dissemination:** Ethical approval is not necessary to be obtained for this systematic review protocol. Results will be disseminated to clinicians and policymakers by publication in a peer-reviewed journal and by presenting at relevant conferences.

**Systematic review registration:** PROSPERO number CRD42021278168

## Strengths and Limitations

- This proposed study will be the most-up-to date systematic review of CIPN ClinROMs and PROMs and the first to use COSMIN methodology to assess the methodological quality of studies reporting psychometric properties of CIPN ClinROMs.
- The proposed study is also the first to use the OMERACT filter 2.1 methodology following COSMIN analysis of CIPN ClinROMs and PROMs. This will facilitate the

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recommendation of one or more ClinROMs and/or PROMs for assessing CIPN, based on the available evidence.

- Subgroup analyses by age group, chemotherapy type, and the timing of the study in relation to the delivery of chemotherapy will be undertaken to gain a better understanding of potential subgroup differences in the psychometric proprieties of CIPN ClinROMs and PROMs.
- Limitations may be related to the heterogeneity of patients who have CIPN and to the lack of consensus about the definition of CIPN.

**Keywords**

COSMIN, Psychometric analysis, chemotherapy-induced peripheral neuropathy, patient-reported outcome measures, clinician-reported outcome measures

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## Introduction

Chemotherapy-induced peripheral neuropathy (CIPN) is a challenging and common side effect of many antineoplastic agents with significant negative impacts on quality of life (1-3). Patients may experience an array of symptoms such as numbness, tingling, pins-and-needles, burning, motor weakness, and/or balance disturbance (1). A meta-analysis has suggested that 68.1% (95% Confidence Interval [CI]: 57.7-78.4) of patients have CIPN symptoms in the first month after completing chemotherapy and 60.0% (95% CI: 36.4-81.6) of patients have symptoms three months after finishing chemotherapy (3, 4). Thirty percent (95% CI: 6.4-53.5) of patients report persistent CIPN symptoms 6 months or later after treatment (4). Wide confidence intervals suggest substantial uncertainty in this estimate, potentially due to variability in measurement tools used in these studies (5, 6). CIPN may lead to dose reductions, delayed treatment or early discontinuation in 2% to 58% of people, with unclear effects on disease status and survival (7-13). Unfortunately, there are no known preventative agents, and only one pharmacologic treatment, duloxetine, with sufficient evidence supporting its use when the outcome is pain intensity (14, 15). However, treatments for the myriad other symptoms associated with CIPN are unavailable.

Early, frequent, and standardized assessment of CIPN is important to understand the development and evolution of symptomatology and to prevent its negative impacts (3, 14). This may be achieved via use of Clinician-Reported Outcome Measures (ClinROM) or Patient-Reported Outcome Measures (PROM). ClinROMs are assessment tools and techniques used by clinicians of observable signs, behaviors, or other manifestations of CIPN (16-20). PROMs allow patients to self-report the presence, intensity, frequency, and/or the impact of symptoms and signs of the sensory, motor, and/or autonomic features of CIPN. CIPN ClinROMs and PROMs must be valid, or measure what they purport to measure(21), reliable (i.e., consistent), reproducible, and stable across raters (22), and responsive to change following administration of the first and subsequent cycles of neurotoxic chemotherapy (20). Unfortunately, there is no agreement on a gold standard CIPN assessment tool, preventing an adequate understanding of the magnitude of the problem of acute and chronic CIPN, its presentation, development and trajectory, risk factors, and impacts (20, 23). Importantly, this may also contribute to difficulty identifying effective preventative agents and treatments (14, 23, 24).

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It is also unclear whether CIPN tools have similar validity and reliability across neurotoxic chemotherapy types, during versus after chemotherapy delivery, or across the adult lifespan (3). For example, while there may be heterogeneity in CIPN experience due to biopsychosocial factors (e.g. genetics, symptom appraisal) (4, 8), CIPN may also differ according to chemotherapy type with potentially different effects on symptom presentation (3). It may also be possible that the psychometric properties of the tools may differ depending on whether CIPN is measured during treatment (e.g. acute symptom presentation) versus after treatment (e.g. chronic symptom presentation). Additionally, aging is accompanied by non-uniform biopsychosocial changes which can influence health and illness (25). In the case of CIPN, age-related morphological changes in the peripheral nervous system, including decreased nerve conduction velocity and slowed axonal regeneration and reinnervation (26-31), may interact with damage to the peripheral nervous system caused by neurotoxic chemotherapy (3), which may also result in differences in symptom presentation. This is important because cancer is primarily a disease of older people (32, 33). Therefore, it may be possible that in the upcoming years, with the aging population (34), there will be many older adults at risk of experiencing CIPN who will require valid and reliable assessment strategies that are able to differentiate normal age-related changes from those associated with CIPN (32, 35).

Although numerous reviews of CIPN ClinROMs and PROMs have been published (6, 36-40), none to date has used methodology that allows for a quantitative assessment and comparison of the methodological quality of psychometric studies or the quality of the CIPN assessment tools, themselves. This is likely a key factor in our inability to identify a gold standard CIPN tool. The COnsensus-based Standards for the selection of health Measurement INstruments (COSMIN) Risk of Bias checklist was developed to quantitatively assess the methodological quality of studies reporting psychometric properties and the ClinROMs and PROMs evaluated in these studies (16, 41, 42). Validity, reliability, and responsiveness are evaluated using 10 boxes that collect information about measurement properties, including measure development, content validity, structural validity, internal consistency, cross-cultural validity/measurement invariance, reliability, measurement error, criterion validity, hypothesis testing, and responsiveness. Assessment criteria in each box depend on the type of analysis performed. The scoring of each question within a box can vary according the measurement propriety assessed (41). The rating scale includes four possible scores, ranging from very good, adequate, doubtful, to inadequate (41). The score

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for each box is calculated based on a "worst score counts" method which only considers the lowest rating within the box. This rigorous method allows for an objective evaluation and quantitative scoring of the quality of studies that report psychometric properties of CIPN ClinROMs and PROMs and the tools themselves. No studies to date have applied the COSMIN methodology to assess the methodological quality of studies reporting psychometric properties of CIPN tools and the overall quality of ClinROMs and/or PROMs.

There are significant gaps in our knowledge about how to best assess CIPN. These knowledge gaps affect our capacity to relieve CIPN symptoms in all adults undergoing neurotoxic chemotherapy. A critical first step to improving our understanding of CIPN across the adult lifespan is to ensure that the assessment tools are valid and reliable. An assessment of the methodological quality of studies reporting psychometric properties of CIPN tools and the quality of ClinROMs and PROMs using the COSMIN method would help to identify gaps and weaknesses in their psychometric properties and potential areas for improvements to existing CIPN measures. Such an analysis would substantially improve the CIPN literature. From a research perspective, it would advance the field toward identification of a gold standard CIPN assessment method, which could have important implications for future trials testing novel preventive and treatment agents. From a clinical perspective, it would provide healthcare providers with valid and reliable assessment tools for early detection and monitoring of CIPN symptoms among younger and older adults undergoing all neurotoxic chemotherapies.

Objectives

The primary objective of this study is to perform a systematic review of studies reporting psychometric properties of CIPN ClinROMs and PROMs among adults receiving, or who had previously received chemotherapy for cancer and to compare COSMIN quality ratings across included studies and the quality of ClinROMs and PROMs, separately. The secondary objective is to perform subgroup analyses of studies by age group (younger adults [18 years old - 59 years old ± 10 years] vs. older adults [≥60 years old ±10 years]), chemotherapy type, and the timing of the study in relation to the delivery of chemotherapy (e.g. during treatment vs after treatment), where data are available.

**Methods and analysis**

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This study protocol was prepared with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols (PRISMA-P) and Preferred Reporting Items for Systematic reviews and Meta-Analyses literature search extension (PRISMA-S) guidelines (43, 44). The PRISMA-P checklist is available in Appendix 1. This systematic review protocol was registered with the International Prospective Register of Systematic Reviews (PROSPERO) on September 13, 2021. Any changes of this protocol will be documented and recorded in PROSPERO. The start date is July 9, 2021, and the anticipated completion date is April 2022.

PICO question

- Population: Adults aged over 18 years receiving or who had previously received chemotherapy for cancer
- Intervention(s), exposure(s): Any type of PROM or ClinROM used to assess CIPN
- Comparator(s)/control: Not applicable
- Main outcome(s): Establishing what is the most appropriate, valid and reliable CIPN PROM and ClinROM using the COSMIN methodology
- Additional outcome: Recommend of one or more ClinROM(s) and/or a PROM(s) for assessing CIPN in clinical and research settings using the OMERACT filter 2.1 methodology.

Eligibility criteria

All studies reporting on the psychometric proprieties of CIPN ClinROMs and/or PROMs among adults aged ≥ 18 years will be eligible for inclusion. There will be no restrictions on publication or year. Articles published in French will be translated to English by bilingual team members (PMB, LRG, DT, MEC) (45). For articles written in other languages we will use Google Translate, as recommended (45). All types of cancer and all types of chemotherapy will be considered. Only empirical, peer-reviewed articles that report on measurement proprieties of CIPN ClinROMs and/or PROMs will be included. Case studies, study protocols, published conference abstracts and systematic reviews will be excluded. Studies reporting data from neonatal and/or pediatric populations combined with data from adult populations will be included if it is possible to isolate the data from the adult subgroup. If it is not possible, these studies will be excluded. Studies will be excluded when an adult proxy is used to assess CIPN symptoms of pediatric patients. Patients who do not have

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cancer or who do not receive any antineoplastic agents will be excluded. Any studies using animals as their population will also be excluded

Information sources:

MEDLINE, Embase, CINAHL, and APA PsycInfo databases was searched from inception to July 8, 2021. Search terms were developed for 1) the population (adults receiving or who had previously received cancer treatment), 2) CIPN, and 3) psychometric properties. The COSMIN search filter for PROMs and ClinROMs was used for the latter conceptual block. Search strategies were verified by a health science librarian with systematic review experience. References of all identified texts will be examined to identify additional studies not identified in the database search.

Search strategy: Search strategies are available in Appendix 2.

Study records: PROSPERO number CRD42021278168

Data management:

All references will be imported into Endnote v20. Duplicates will be removed using Bramer's de-duplication method in Endnote (46). Then, references will be exported into Covidence in order to carry out the study selection (47).

Study Selection process:

A two-stage process for study screening will be used. Two independent raters (PBM, LAR) will screen titles and abstracts of articles to assess whether they meet the eligibility criteria. Full-text screening will be conducted by the 2 independent raters (PBM, KM). The number of included and excluded studies at each step will be presented in a PRISMA flow chart. Reasons for the exclusion of full-text articles will be recorded in Covidence and presented in the PRISMA flow chart. A third rater (LRG) will resolve all disagreements at each screening phase. Percentage agreement between raters will be calculated at each phase to report interrater reliability (48).

Data Collection Process

For each study included in the review, data relating to study characteristics (target population, mode of administration, recall period, subscale(s) and number of items,

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response options, range of scores, original language and available translation) and measurement properties of the PROMs and the ClinROMs (instrument development, content validity, structural validity, internal consistency, cross-cultural validity/measurement invariance, reliability, measurement error, criterion validity, hypothesis testing for the construct validity and responsiveness) will be extracted by 2 people and compared for consistency (PBM, MEC). CIPN definitions reported by the authors will also be extracted to examine and compare conceptual definitions across studies. Data extraction tables will be adapted from those designed by the COSMIN group. Corresponding authors will be contacted in the event of missing data (descriptive data, missing information about the items and the scoring of each tool). Three contact attempts will be made over the period of one month. In the case of non-response, data will be coded as missing.

Quality Assessment

The COSMIN Risk of Bias checklist will be used by two independent raters (PBM, MEC) to assess the methodological quality of the included studies and the quality of ClinROMs and PROMs (42). The first step consists of assessing the quality of each study describing psychometric properties of CIPN ClinROMs and PROMs using an Excel spreadsheet proposed by the COSMIN group (49). In this step, each of the 10 measurement properties considered by the COSMIN system is given 4 possible scores: very good, adequate, doubtful and inadequate (49). A summary table which presents the quality rating of each study will be included in this review once the COSMIN Risk of Bias checklist is completed. The second step involves rating each study describing psychometric properties of CIPN ClinROMs and PROMs against the criteria for good measurement properties (49). Each rating has a possible score of sufficient (+), insufficient (-), inconsistent (±) or indeterminate rating (?) according to published criteria (49). The third step involves determining an overall quality score of all studies reporting on a given ClinROM or PROM by considering cross-study ratings (49). The quality of evidence is determined by using a modified GRADE approach with a possibility of 4 grades: high, moderate, low, very low (49). Psychometric data from ClinROMs will be considered separately from psychometric data from PROMs. We will rely on data for steps 1 and 2 from a recently completed systematic review and COSMIN analysis of CIPN PROMs (PROSPERO 2020 CRD42020210405; Li, Park & Rutherford, Manuscript in progress) for any studies overlapping across the 2 reviews.

Subgroup analysis

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The COSMIN Risk of Bias checklist will be completed and analyzed to compare methodological quality ratings from studies reporting psychometric properties of CIPN tools and the quality of ClinROMs and PROMs across age groups (younger adults [18 years old - 59 years old ± 10 years] vs. older adults [≥60 years old ±10 years]), the type of chemotherapy received, and the timing of the study in relation to the delivery of chemotherapy (e.g. during treatment vs after treatment), where data are available. Age subgroups are based on cut-offs used in previous studies in cancer with clinically relevant outcomes (50-52). We will allow for the cutoff between younger and older groups to vary by 20 years to account for methodological differences in the categorization of younger and older age groups across studies.

Recommendation of a ClinROM and PROM

While the COSMIN method is the only available method to quantitatively assess the methodological quality of studies reporting psychometric properties and the ClinROMs and PROMs, it does not provide a method to recommend the use of a ClinROMs or PROMs. In this review, we will address this limitation by using the Outcome Measures in Rheumatology (OMERACT) filter 2.1, based on the findings of the COSMIN analysis, to determine whether it is possible to recommend the use of one or more ClinROMs or PROMs, and if so, make recommendations (53, 54). Data issued by COSMIN on construct validity, test-retest reliability and longitudinal construct validity (i.e. responsiveness) and additional information on clinical trial discrimination and thresholds of meaning which are not covered in COSMIN will be used to complete the COSMIN-OMERACT Good Measures Checklist by two independent raters (53). This methodology permits recommendations based on a standardized, rigorous and transparent process (53). Although this methodology has not been used in systematic reviews of the psychometric properties of CIPN measures, 2 previous reviews have used the OMERACT filter 2.1 in conjunction with the results of a COSMIN analysis to make recommendations about dermatological measures (55, 56).

The OMERACT filter 2.1 (Table 1) has 3 pillars, 4 questions, 7 measurement properties and 1 answer (53). Truth, discrimination, and feasibility represent the pillars (53). A set of questions are associated with each pillar which are organized in an algorithm (53). Each question has a possibility of 4 scores: Red, amber, green, and white (53). Red means “stop, do not continue”, amber means “a caution is raised but you can continue”, green means “go, this question is definitely answered affirmatively”, and white means an absence of evidence

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and the evaluation must stop (53). An overall score that ranges between do not endorse, provisionally endorse, or endorse is issued once the 4 questions of the algorithm are answered (53). The 7 included measurement properties map onto 7 of the COSMIN measurement properties. We will adapt the OMERACT filter 2.1 to include all 10 COSMIN Risk of bias checklist measurement properties.

**Table 1 OMERACT filter 2.1 definitions, questions, and measurement properties**

| Pillars | Definition | Questions | Measurement properties |
|---|---|---|---|
| Truth | The ability of the outcome measurement tool to measure what is intended (53) | "Is it a match with the target domain?" | Construct validity, reliability |
| | | "Do the numeric scores make sense?" | Content validity, face validity |
| Discrimination | The ability of the outcome measurement tool to discriminate different situations of interest (53) | "Can it discriminate between groups of interest?" | Test-retest reliability, longitudinal validity/responsiveness, ability to discriminate in RCT/comparative research setting, threshold of meaning |
| Feasibility | The practicality of the outcome measurement tool (time, cost, burden) (53) | "Is it practical to use?" | Access, training, translation, length, cost, burden |

Data synthesis and best-evidence synthesis

The Synthesis Without Meta-analysis (SWiM) guideline will be used to guide data synthesis for the included studies (57). SWiM includes 9 items that report key features in the methods, results and discussion of every study, such as grouping studies for synthesis (item 1), standardized metric used for synthesis (item 2), synthesis methods and their limitations (items 3 and 9), criteria used to prioritize results for summary and synthesis (item 4),

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heterogeneity in reported effects (item 5), certainty of evidence (item 6), data presentation methods (item 7) and a summary of the synthesis (item 8) (57).

Patient and public involvement

A patient partner and co-author (MB) participated in the development of the review protocol and will participate as a research team member throughout the review process, contributing to interpretation of the findings, manuscript drafting and revisions.

## Ethics and dissemination

An ethical approval is not necessary to be obtained for this systematic review protocol. The results will be disseminated to clinicians and researchers by publication in a peer-reviewed journal and by presenting at relevant conferences. This systematic review will support researchers and clinicians to use the best measure to assess CIPN.

## Discussion

This proposed study will be the most-up-to date systematic review of CIPN ClinROMs and PROMs and the only one to use the COSMIN risk of bias tool with CIPN ClinROMs, allowing for a quantitative evaluation and comparison of the methodological quality of studies reporting psychometric properties of CIPN tools and the quality of ClinROMs and PROMs. Subgroup analyses by age group, chemotherapy type, and the timing of the study in relation to the delivery of chemotherapy will be carried out to determine whether the psychometric proprieties of CIPN ClinROMs and PROMs are the same across clinically relevant subgroups. This systematic review is the first to use the OMERACT filter 2.1 methodology. This will facilitate the recommendation of one or more ClinROMs and/or a PROMs for assessing CIPN in clinical and research settings, based on the available evidence.

## Declaration

Competing interest statement

The authors declare that they have no competing interests

Funding statement

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## Authors' Contributions

PBM and LRG conceived the study idea and were responsible for developing and writing the first draft of the systematic review protocol and manuscript. EB and PBM developed search routine for each database. TL, SP, LAR, MEC, KM, DT, MB and JG provided critical insights at all stages. All authors approved and contributed to the final manuscript.

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**Appendix 1: PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol***

| Section and topic | Item No | Checklist item | Page number |
|---|---|---|---|
| **ADMINISTRATIVE INFORMATION** | | | |
| Title: | | | |
| Identification | 1a | Identify the report as a protocol of a systematic review | p.1 |
| Update | 1b | If the protocol is for an update of a previous systematic review, identify as such | n/a |
| Registration | 2 | If registered, provide the name of the registry (such as PROSPERO) and registration number | pp.2,6,8 |
| Authors: | | | |
| Contact | 3a | Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author | p.1 |
| Contributions | 3b | Describe contributions of protocol authors and identify the guarantor of the review | p.12 |
| Amendments | 4 | If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments | n/a |
| Support: | | | |
| Sources | 5a | Indicate sources of financial or other support for the review | p.12 |
| Sponsor | 5b | Provide name for the review funder and/or sponsor | p.12 |
| Role of sponsor or funder | 5c | Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol | p.12 |
| **INTRODUCTION** | | | |
| Rationale | 6 | Describe the rationale for the review in the context of what is already known | pp.4-6 |
| Objectives | 7 | Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO) | p.6 |
| **METHODS** | | | |
| Eligibility criteria | 8 | Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review | p.7 |

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| | | | |
|---|---|---|---|
| Information sources | 9 | Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage | pp.7-8 |
| Search strategy | 10 | Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated | pp.19-32 |
| Study records: | | | |
| Data management | 11a | Describe the mechanism(s) that will be used to manage records and data throughout the review | p.8 |
| Selection process | 11b | State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis) | p.8 |
| Data collection process | 11c | Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators | pp.8-9 |
| Data items | 12 | List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications | pp.6, 8 |
| Outcomes and prioritization | 13 | List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale | p.6 |
| Risk of bias in individual studies | 14 | Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis | p.9 |
| Data synthesis | 15a | Describe criteria under which study data will be quantitatively synthesised | p.11 |
| | 15b | If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as $I^2$, Kendall's $\tau$) | p.11 |
| | 15c | Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression) | p.9 |
| | 15d | If quantitative synthesis is not appropriate, describe the type of summary planned | pp.9-11 |
| Meta-bias(es) | 16 | Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies) | n/a |
| Confidence in cumulative evidence | 17 | Describe how the strength of the body of evidence will be assessed (such as GRADE) | n/a |

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

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*From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647*

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## Appendix 2: Search strategies

### Medline (OVID)

| Psychometrics properties | | Results |
|---|---|---|
| **1** | instrumentation.fs OR validation study.pt OR exp reproducibility of results/ OR psychometrics/ OR observer variation/ OR discriminant analysis/ | |
| **2** | (reproducib* OR psychometr* OR clinimetr* OR clinometr* OR observer variation OR reliab* OR valid* OR coefficient OR internal consistency OR (cronbach* AND (alpha OR alphas)) OR item correlation* OR item selection* OR item reduction*).**ti,ab** OR (agreement OR precision OR imprecision OR precise values).**tw** OR (test retest OR (test AND retest) OR (reliab* AND (test OR retest)) OR stability OR interrater OR inter rater OR intrarater OR intra rater OR intertester OR inter tester OR intratester OR intra tester OR interobserver OR inter observer OR intraobserver OR intra observer OR intertechnician OR inter technician OR intratechnician OR intra technician OR interexaminer OR inter examiner OR intraexaminer OR intra examiner OR interassay OR inter assay OR intraassay OR intra assay OR interindividual OR inter individual OR intraindividual OR intra individual OR interparticipant OR inter participant OR intraparticipant OR intra participant OR kappa OR kappa's OR kappas OR coefficient of variation).**ti,ab**<br>OR (repeatab* OR ((replicab* OR repeated) AND (measure OR measures OR findings OR result OR results OR test OR tests))).**tw**<br>OR (generaliza* OR generalisa* OR concordance OR (intraclass AND correlation*) OR discriminative OR known group OR factor analysis OR factor analyses OR factor structure OR factor structures OR dimensionality OR subscale* OR multitraitscaling analysis OR multitrait scaling analyses OR item discriminant OR interscale correlation* OR ((error OR errors) AND (measure*OR correlat* OR evaluat* OR accuracy OR accurate OR precision OR mean)) OR individual variability OR interval variability OR rate variability OR variability analysis OR (uncertainty AND (measurement OR measuring)) OR standard error of measurement OR sensitiv* OR responsive* OR (limit AND detection) OR minimal detectable concentration OR interpretab* OR (small* AND (real OR detectable) AND (change OR difference)) OR meaningful change OR minimal important change OR minimal important difference OR minimally important change OR minimally important difference OR minimal detectable change OR minimal detectable difference OR minimally detectable change OR minimally detectable difference OR minimal real change OR minimal real difference OR minimally real change OR minimally real difference OR ceiling effect OR floor effect OR Item response model OR IRT OR Rasch OR Differential item functioning OR DIF OR computer adaptive testing OR itembank OR cross cultural equivalence).**ti,ab** | |
| **3** | **1 OR 2** | **4108782** |

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| | | |
|---|---|---|
| **Patient-reported outcome measurement instruments** | | |
| **4** | "patient reported outcome measures"/ or "surveys and questionnaires"/ or self report/ or quality of life/ | |
| **5** | (((patient? OR self) **adj2** (report* OR relate* OR declar* OR rating* OR assess* OR examin* OR monitor* OR administer* OR rate* OR rating* or based or appraisal* or appraised or evaluat*) **adj2** (outcome* OR symptom*)) or PROM? Or PRO or PROS OR (quality adj2 life) OR HRQL or HRQoL or QL or QoL OR HR-PRO or HRPRO).**ti,ab,kw** | |
| **6** | ((disability or function* or subjective or utility or utilities or wellbeing or well being) **adj3** (outcome or outcomes or index or indices or instrument or instruments or measure or measures or questionnaire or questionnaires or profile or profiles or scale or scales or score or scores or scoring or status or survey or surveys or measure*)).**ti,ab,kw** | |
| **Clinician reported outcome measurement instruments** | | |
| **7** | exp outcome assessment health care/ | |
| **8** | (Clinician* reported outcome* or CROS or clinRO or clinROs OR ((clinical or clinician* or expert* or health professional* OR physician* OR doctor*) **adj2** (judgement* or interpretation* or observation* or assessment* or measure*))).**ti,ab** | |
| **9** | **OR/4-8** | **2310354** |
| **Peripheral neuropathy** | | |
| **10** | peripheral nervous system diseases/ or brachial plexus neuropathies/ or exp complex regional pain syndromes/ or neuralgia/ or metatarsalgia/ or neuritis/ or polyneuropathies/ or polyradiculoneuropathy/ or small fiber neuropathy/ or neural conduction/ or cranial nerve injuries/ or peripheral nerve injuries/ | |
| **11** | (PNS disease* or peripheral nervous system disease* or neuropath* or complex regional pain syndrome* or neuralgia* or neuritis or neurodinya* or (nerve* **adj2** (pain* or injur* or disease* or affect* or damage* or degeneration*)) or causalgia* or hyperesthesi* or paresthesi* or hypesthesi* or paralysis or polyneuropath* or polyradiculo* or polyneuritis or burning feet syndrome or metatarsalgia).**ti,ab** | |
| **12** | neurologic manifestations/ | |
| **13** | ((neurologic* **adj2** (manifestation* or deficit* or sign or signs or symptom* or finding* or dysfunction*))).**ti,ab** | |
| **14** | sensation disorders/ or exp somatosensory disorders/ or sensory thresholds/ or differential threshold/ or pain threshold/ or signal detection, psychological/ or "evoked potentials, somatosensory"/ or touch perception/ or exp nociceptive pain/ or pain, referred/ or exp pain perception/ or breakthrough pain/ or facial pain/ or neurotoxicity syndromes/ or hand-foot syndrome/ | |
| **15** | (((sensation* or sense* or sensivity or sensorimotor or sensory or somatosensory or nociceptive or somatic or tissue* or tactile or touch* or pinprick or thermal or burning or shooting or deafferentation or differential or referred or percept* or facial or breakthrought) **adj3** (pain* or disorder* or disease* or disturb*or distortion* or | |

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| | | |
|---|---|---|
| | discomfort* or dysfunction* or disruption* or impair* or symptom* or change* or abnormal* or mechanical or deficit* or limit* or loss or diminish* or reduc* or amplificat* or hyperesthetic or alter* or affect* or gating or threshold*)) or pain threshold* or hyperalgesi* or allodynia* or formication* or tingling or hypersensitivity or oxyesthesi* or electric shock like or vibration* or nocicept* or neurotoxi* or hand foot syndrome*).**ti,ab** | |
| **16** | exp "musculoskeletal and neural physiological phenomena"/ or muscular diseases/ or neuromuscular manifestations/ or muscle weakness/ or muscle cramp/ or arthralgia/ or exp gait disorders, neurologic/ or proprioception/ or weight-bearing/ or evoked potentials, motor/ | |
| **17** | ((motor* or muscle* or muscular or neuromuscular or functional* or movement*or walk* or gait* or balance or strength or tendon*or joint* or reflex* or tonus) **adj3** (pain* or disorder* or disturb* or discomfort* or dysfunction* or disruption* or impair* or symptom* or change* or abnormal* or mechanical or deficit* or limit* or loss or diminish* or reduc* or alter* or disabilit* or postural or weak* or fatigue) or propriocepti* or arthralgia* or myalgia* or polyarthralgia* or numbness or foot drop or footdrop or cramp*)**.ti,ab.** | |
| **18** | **OR/10-17** | **2852840** |
| **CIPN** | | |
| **19** | ((chemotherapy induced adj2 neuropath*) or CIPN).**ti,ab** | |
| **Antineoplastic agents** | | |
| **20** | Exp antineoplastic agents/ or antineoplastic combined chemotherapy protocols/ | |
| **21** | (antineoplastic* or chemotherap* or ((anticancer or antitumo?) **adj3** (agent or agents or drug*))).**ti,ab** | |
| **22** | **20 OR 21** | **1364658** |
| **23** | **3 AND 9 AND 18 AND 22** | **2626** |
| **24** | **3 AND 9 AND 19** | **151** |
| **25** | **23 OR 24** | **2626** |
| **Exclusion Filters** | | |
| **26** | (address OR case reports OR comment OR directory OR editorial OR festschrift OR lectures OR legal cases OR legislation OR letter OR news OR newspaper article OR patient education handout OR popular work OR congress OR practice guideline).**pt** | |
| **27** | exp biography/ OR exp consensus development conference/ | |
| **28** | **26 or 27** | **4139201** |
| **29** | **25 NOT 28** | **2450** |

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**Embase (OVID)**

| | **Psychometrics properties** | **Results** |
|---|---|---|
| **1** | devices/ OR validation study/ OR reproducibility/ OR exp psychometry/ or observer variation/ OR-discriminant analysis/ OR validity/ | |
| **2** | (reproducib* OR psychometr* OR clinimetr* OR clinometr* OR observer variation OR reliab* OR valid* OR coefficient OR internal consistency OR (cronbach* AND (alpha OR alphas)) OR item correlation* OR item selection* OR item reduction*).**ti,ab** OR (agreement OR precision OR imprecision OR precise values).**tw** OR (test retest OR (test AND retest) OR (reliab* AND (test OR retest)) OR stability OR interrater OR inter rater OR intrarater OR intra rater OR intertester OR inter tester OR intratester OR intra tester OR interobserver OR inter observer OR intraobserver OR intra observer OR intertechnician OR inter technician OR intratechnician OR intra technician OR interexaminer OR inter examiner OR intraexaminer OR intra examiner OR interassay OR inter assay OR intraassay OR intra assay OR interindividual OR inter individual OR intraindividual OR intra individual OR interparticipant OR inter participant OR intraparticipant OR intra participant OR kappa OR kappa's OR kappas OR coefficient of variation).**ti,ab**<br>OR (repeatab* OR ((replicab* OR repeated) AND (measure OR measures OR findings OR result OR results OR test OR tests))).**tw**<br>OR (generaliza* OR generalisa* OR concordance OR (intraclass AND correlation*) OR discriminative OR known group OR factor analysis OR factor analyses OR factor structure OR factor structures OR dimensionality OR subscale* OR multitraitscaling analysis OR multitrait scaling analyses OR item discriminant OR interscale correlation* OR ((error OR errors) AND (measure*OR correlat* OR evaluat* OR accuracy OR accurate OR precision OR mean)) OR individual variability OR interval variability OR rate variability OR variability analysis OR (uncertainty AND (measurement OR measuring)) OR standard error of measurement OR sensitiv* OR responsive* OR (limit AND detection) OR minimal detectable concentration OR interpretab* OR (small* AND (real OR detectable) AND (change OR difference)) OR meaningful change OR minimal important change OR minimal important difference OR minimally important change OR minimally important difference OR minimal detectable change OR minimal detectable difference OR minimally detectable change OR minimally detectable difference OR minimal real change OR minimal real difference OR minimally real change OR minimally real difference OR ceiling effect OR floor effect OR Item response model OR IRT OR Rasch OR Differential item functioning OR DIF OR computer adaptive testing OR itembank OR cross cultural equivalence).**ti,ab** | |
| **3** | **1 OR 2** | **5454804** |
| **Patient-Reported outcome measurement** | | |
| **4** | patient-reported outcome measure/ OR questionnaire/ OR self report/ OR quality of life/ | |

| | | |
|---|---|---|
| **5** | (((patient? OR self) **adj2** (report* OR relate* OR declar* OR rating* OR assess* OR examin* OR monitor* OR administer* OR rate* OR rating* or based or appraisal* or appraised or evaluat*) **adj2** (outcome* OR symptom*) or PROM? Or PRO or PROS OR (quality adj2 life) OR HRQL or HRQoL or QL or QoL OR HR-PRO or HRPRO).**ti,ab,kw** | |
| **6** | ((disability or function* or subjective or utility or utilities or wellbeing or well being) adj3 (outcome or outcomes or index or indices or instrument or instruments or measure or measures or questionnaire or questionnaires or profile or profiles or scale or scales or score or scores or scoring or status or survey or surveys or measure*)).**ti,ab,kw** | |
| **Clinician reported outcomes measurement** | | |
| **7** | exp outcome assessment/ | |
| **8** | (Clinician* reported outcome* or CROS or clinRO or clinROs OR ((clinical or clinician* or expert* or health professional* OR physician* OR doctor*) **adj2** (judgement* or interpretation* or observation* or assessment* or measure*))).**ti,ab** | |
| **9** | **OR/4-8** | **2698996** |
| **Peripheral neuropathy** | | |
| **10** | peripheral neuropathy/ OR brachial plexus neuropathy/ OR exp complex regional pain syndrome/ OR neuralgia/ OR metatarsalgia/ OR neuritis/ OR polyneuropathy/ OR polyradiculoneuropathy/ OR small fiber neuropathy/ OR nerve conduction/ OR cranial nerve injury/ OR peripheral nerve injury/ | |
| **11** | (PNS disease* or peripheral nervous system disease* or neuropath* or complex regional pain syndrome* or neuralgia* or neuritis or neurodinya* or (nerve* **adj2** (pain* or injur* or disease* or affect* or damage* or degeneration*)) or causalgia* or hyperesthesi* or paresthesi* or hypesthesi* or paralysis or polyneuropath* or polyradiculo* or polyneuritis or burning feet syndrome or metatarsalgia).**ti,ab** | |
| **12** | neurologic disease/ | |
| **13** | ((neurologic* **adj2** (manifestation* or deficit* or sign or signs or symptom* or finding* or dysfunction*))).**ti,ab** | |
| **14** | sensory dysfunction/ OR exp somatosensory disorder/ OR perceptive threshold/ OR differential threshold/ OR pain threshold/ OR signal detection/ OR somatosensory evoked potential/ OR touch/ OR exp nociceptive pain/ OR referred pain/ OR exp nociception/ OR breakthrough pain/ OR face pain/ OR neurotoxicity/ OR hand foot syndrome/ | |
| **15** | (((sensation* or sense* or sensivity or sensorimotor or sensory or somatosensory or nociceptive or somatic or tissue* or tactile or touch* or pinprick or thermal or burning or shooting or deafferentation or differential or referred or percept* or facial or breakthrought) **adj3** (pain* or disorder* or disease* or disturb*or distortion* or discomfort* or dysfunction* or disruption* or impair* or symptom* or change* or abnormal* or mechanical or deficit* or limit* or loss or diminish* or reduc* or amplificat* or hyperesthetic or alter* or affect* or gating or | |

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| | | |
|---|---|---|
| | threshold*)) or pain threshold* or hyperalgesi* or allodynia* or formication* or tingling or hypersensitivity or oxyesthesi* or electric shock like or vibration* or nocicept* or neurotoxi* or hand foot syndrome*).**ti,ab** | |
| **16** | "biological phenomena and functions concerning organ systems"/ OR muscle disease/ OR neuromuscular disease/ OR muscle weakness/ OR muscle cramp/ OR arthralgia/ OR exp neurologic gait disorder/ OR proprioception/ OR weight bearing/ OR motor evoked potential/ | |
| **17** | ((motor* or muscle* or muscular or neuromuscular or functional* or movement*or walk* or gait* or balance or strength or tendon*or joint* or reflex* or tonus) **adj3** (pain* or disorder* or disturb* or discomfort* or dysfunction* or disruption* or impair* or symptom* or change* or abnormal* or mechanical or deficit* or limit* or loss or diminish* or reduc* or alter* or disabilit* or postural or weak* or fatigue) or propriocepti* or arthralgia* or myalgia* or polyarthralgia* or numbness or foot drop or footdrop or cramp*).**ti,ab**. | |
| **18** | **OR/10-17** | **2076319** |
| **CIPN** | | |
| **19** | chemotherapy-induced peripheral neuropathy/ | |
| **20** | ((chemotherapy induced **adj2** neuropath*) or CIPN).**ti,ab** | |
| **21** | **19 OR 20** | **2643** |
| **Antineoplastic agents** | | |
| **22** | exp antineoplastic agent/ OR combination chemotherapy/ | |
| **23** | (antineoplastic* or chemotherap* or ((anticancer or antitumo?) **adj3** (agent or agents or drug*))).**ti,ab** | |
| **24** | **22 OR 23** | **2693841** |
| **25** | **3 AND 9 AND 18 AND 24** | **5455** |
| **26** | **3 AND 9 AND 21** | **394** |
| **27** | **25 OR 26** | **5458** |
| **Exclusion Filters** | | |
| **28** | (editorial or letter).pt | |
| **29** | literature/ or case report/ or directory/ or editorial/ or interview/ or law/ or publication/ or consensus development/ or practice guideline/ | |
| **30** | **28 OR 29** | **5146595** |
| **31** | **27 NOT 30** | **4891** |

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**CINAHL:**

| Psychometrics properties | | Results |
|---|---|---|
| **1** | **MH**("Instrument Construction+" OR "Validation Studies" OR "Reproducibility of Results+" OR "Psychometrics+" OR "Observer Bias+" OR "Discriminant Analysis+") | |
| **2** | **TI**(reproducib* OR psychometr* OR clinimetr* OR clinometr* OR observer variation OR reliab* OR valid* OR coefficient OR internal consistency OR (cronbach* AND (alpha OR alphas)) OR item correlation* OR item selection* OR item reduction* OR agreement OR precision OR imprecision OR precise values OR test retest OR (test AND retest) OR (reliab* AND (test OR retest)) OR stability OR interrater OR inter rater OR intrarater OR intra rater OR intertester OR inter tester OR intratester OR intra tester OR interobserver OR inter observer OR intraobserver OR intra observer OR intertechnician OR inter technician OR intratechnician OR intra technician OR interexaminer OR inter examiner OR intraexaminer OR intra examiner OR interassay OR inter assay OR intraassay OR intra assay OR interindividual OR inter individual OR intraindividual OR intra individual OR interparticipant OR inter participant OR intraparticipant OR intra participant OR kappa OR kappa's OR kappas OR coefficient of variation OR repeatab* OR ((replicab* OR repeated) AND (measure OR measures OR findings OR result OR results OR test OR tests)) OR generaliza* OR generalisa* OR concordance OR (intraclass AND correlation*) OR discriminative OR known group OR factor analysis OR factor analyses OR factor structure OR factor structures OR dimensionality OR subscale* OR multitraitscaling analysis OR multitrait scaling analyses OR item discriminant OR interscale correlation* OR ((error OR errors) AND (measure*OR correlat* OR evaluat* OR accuracy OR accurate OR precision OR mean)) OR individual variability OR interval variability OR rate variability OR variability analysis OR (uncertainty AND (measurement OR measuring)) OR standard error of measurement OR sensitiv* OR responsive* OR (limit AND detection) OR minimal detectable concentration OR interpretab* OR (small* AND (real OR detectable) AND (change OR difference)) OR meaningful change OR minimal important change OR minimal important difference OR minimally important change OR minimally important difference OR minimal detectable change OR minimal detectable difference OR minimally detectable change OR minimally detectable difference OR minimal real change OR minimal real difference OR minimally real change OR minimally real difference OR ceiling effect OR floor effect OR Item response model OR IRT OR Rasch OR Differential item functioning OR DIF OR computer adaptive testing OR itembank OR cross cultural equivalence) | |
| **3** | **AB(**reproducib* OR psychometr* OR clinimetr* OR clinometr* OR observer variation OR reliab* OR valid* OR coefficient OR internal consistency OR (cronbach* AND (alpha OR alphas)) OR item correlation* OR item selection* OR item reduction* OR agreement OR precision OR imprecision OR precise values OR test retest OR (test AND retest) OR (reliab* AND (test OR retest)) OR stability OR interrater OR inter rater OR | |

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| | intrarater OR intra rater OR intertester OR inter tester OR intratester OR intra tester OR interobserver OR inter observer OR intraobserver OR intra observer OR intertechnician OR inter technician OR intratechnician OR intra technician OR interexaminer OR inter examiner OR intraexaminer OR intra examiner OR interassay OR inter assay OR intraassay OR intra assay OR interindividual OR inter individual OR intraindividual OR intra individual OR interparticipant OR inter participant OR intraparticipant OR intra participant OR kappa OR kappa's OR kappas OR coefficient of variation OR repeatab* OR ((replicab* OR repeated) AND (measure OR measures OR findings OR result OR results OR test OR tests)) OR generaliza* OR generalisa* OR concordance OR (intraclass AND correlation*) OR discriminative OR known group OR factor analysis OR factor analyses OR factor structure OR factor structures OR dimensionality OR subscale* OR multitraitscaling analysis OR multitrait scaling analyses OR item discriminant OR interscale correlation* OR ((error OR errors) AND (measure*OR correlat* OR evaluat* OR accuracy OR accurate OR precision OR mean)) OR individual variability OR interval variability OR rate variability OR variability analysis OR (uncertainty AND (measurement OR measuring)) OR standard error of measurement OR sensitiv* OR responsive* OR (limit AND detection) OR minimal detectable concentration OR interpretab* OR (small* AND (real OR detectable) AND (change OR difference)) OR meaningful change OR minimal important change OR minimal important difference OR minimally important change OR minimally important difference OR minimal detectable change OR minimal detectable difference OR minimally detectable change OR minimally detectable difference OR minimal real change OR minimal real difference OR minimally real change OR minimally real difference OR ceiling effect OR floor effect OR Item response model OR IRT OR Rasch OR Differential item functioning OR DIF OR computer adaptive testing OR itembank OR cross cultural equivalence) | |
| **4** | **1 OR 2 OR 3** | **40141** |
| **Patient-Reported Outcome Measurements** | | |
| **5** | **MH**("Patient-Reported Outcomes+" OR "Surveys+" OR "Questionnaires+ OR "Self Report+" OR "Quality of Life+") | |
| **6** | **TI((**(patient# OR self) **N2** (report* OR relate* OR declar* OR rating* OR assess* OR examin* OR monitor* OR administer* OR rate* OR rating* or based or appraisal* or appraised or evaluat*) **N2** (outcome* OR symptom*)) or PROM* Or PRO or PROS OR (quality **N2** life) OR HRQL or HRQoL or QoL OR HR PRO or HRPRO) | |
| **7** | **AB**(((patient# OR self) **N2** (report* OR relate* OR declar* OR rating* OR assess* OR examin* OR monitor* OR administer* OR rate* OR rating* or based or appraisal* or appraised or evaluat*) **N2** (outcome* OR symptom*)) or PROM* Or PRO or PROS OR (quality **N2** life) OR HRQL or HRQoL or QoL OR HR PRO or HRPRO) | |

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| | | |
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| **8** | **TI**((disability or function* or subjective or utility or utilities or wellbeing or well being) **N3** (outcome or outcomes or index or indices or instrument or instruments or measure or measures or questionnaire or questionnaires or profile or profiles or scale or scales or score or scores or scoring or status or survey or surveys or measure*)) | |
| **9** | **AB**((disability or function* or subjective or utility or utilities or wellbeing or well being) **N3** (outcome or outcomes or index or indices or instrument or instruments or measure or measures or questionnaire or questionnaires or profile or profiles or scale or scales or score or scores or scoring or status or survey or surveys or measure*)) | |
| **Clinician reported outcomes measurements** | | |
| **10** | (**MH** "Outcome Assessment") | |
| **11** | **TI**(Clinician* reported outcome* or CROS or clinRO or clinROs OR ((clinical or clinician* or expert* or health professional* OR physician* OR doctor*) **N2** (judgement* or interpretation* or observation* or assessment* or measure*))) | |
| **12** | **AB**(Clinician* reported outcome* or CROS or clinRO or clinROs OR ((clinical or clinician* or expert* or health professional* OR physician* OR doctor*) **N2** (judgement* or interpretation* or observation* or assessment* or measure*))) | |
| **13** | **OR/5-12** | **908267** |
| **Peripheral neuropathy** | | |
| **14** | **MH**(peripheral nervous system diseases or brachial plexus neuropathies+ or complex regional pain syndrome or neuralgia or metatarsalgia or neuritis or polyneuropathies or polyradiculopathy or small fiber neuropathy or neural conduction or peripheral nerves+) | |
| **15** | **TI**(PNS disease* or peripheral nervous system disease* or neuropath* or complex regional pain syndrome* or neuralgia* or neuritis or neurodinya* or (nerve* **N2** (pain* or injur* or disease* or affect* or damage* or degeneration*)) or causalgia* or hyperesthesi* or paresthesi* or hypesthesi* or paralysis or polyneuropath* or polyradiculo* or polyneuritis or burning feet syndrome or metatarsalgia) | |
| **16** | **AB**(PNS disease* or peripheral nervous system disease* or neuropath* or complex regional pain syndrome* or neuralgia* or neuritis or neurodinya* or (nerve* **N2** (pain* or injur* or disease* or affect* or damage* or degeneration*)) or causalgia* or hyperesthesi* or paresthesi* or hypesthesi* or paralysis or polyneuropath* or polyradiculo* or polyneuritis or burning feet syndrome or metatarsalgia) | |
| **17** | **MH**(neurologic manifestations) | |
| **18** | TI((neurologic* **N2** (manifestation* or deficit* or sign or signs or symptom* or finding* or dysfunction*))) | |
| **19** | AB((neurologic* **N2** (manifestation* or deficit* or sign or signs or symptom* or finding* or dysfunction*))) | |

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| **20** | MH (sensation disorders+ or sensation+ or evoked potentials+ or nociceptive pain+ or breakthrough pain or facial pain or referred pain or neurotoxicity syndromes or hand-foot syndrome) | |
| **21** | **TI**(((sensation* or sense* or sensivity or sensorimotor or sensory or somatosensory or nociceptive or somatic or tissue* or tactile or touch* or pinprick or thermal or burning or shooting or deafferentation or differential or referred or percept* or facial or breakthrought) **N3** (pain* or disorder* or disease* or disturb*or distortion* or discomfort* or dysfunction* or disruption* or impair* or symptom* or change* or abnormal* or mechanical or deficit* or limit* or loss or diminish* or reduc* or amplificat* or hyperesthetic or alter* or affect* or gating or threshold*)) or pain threshold* or hyperalgesi* or allodynia* or formication* or tingling or hypersensitivity or oxyesthesi* or electric shock like or vibration* or nocicept* or neurotoxi* or hand foot syndrome*) | |
| **22** | **AB**(((sensation* or sense* or sensivity or sensorimotor or sensory or somatosensory or nociceptive or somatic or tissue* or tactile or touch* or pinprick or thermal or burning or shooting or deafferentation or differential or referred or percept* or facial or breakthrought) **N3** (pain* or disorder* or disease* or disturb*or distortion* or discomfort* or dysfunction* or disruption* or impair* or symptom* or change* or abnormal* or mechanical or deficit* or limit* or loss or diminish* or reduc* or amplificat* or hyperesthetic or alter* or affect* or gating or threshold*)) or pain threshold* or hyperalgesi* or allodynia* or formication* or tingling or hypersensitivity or oxyesthesi* or electric shock like or vibration* or nocicept* or neurotoxi* or hand foot syndrome*) | |
| **23** | **MH**(musculoskeletal, neural, and ocular physiology or muscle fatigue or muscle strength+ or muscle tonus or muscular diseases or neuromuscular manifestations or reflex or muscle weakness or muscle cramp or arthralgia or gait disorders, neurologic+ or motor activity or physical mobility or muscle pain or neuromuscular control or weight-bearing or weight shifting) | |
| **24** | **TI**((motor* or muscle* or muscular or neuromuscular or functional* or movement*or walk* or gait* or balance or strength or tendon*or joint* or reflex* or tonus) **N3** (pain* or disorder* or disturb* or discomfort* or dysfunction* or disruption* or impair* or symptom* or change* or abnormal* or mechanical or deficit* or limit* or loss or diminish* or reduc* or alter* or disabilit* or postural or weak* or fatigue) or propriocepti* or arthralgia* or myalgia* or polyarthralgia* or numbness or foot drop or footdrop or cramp*) | |
| **25** | **AB**((motor* or muscle* or muscular or neuromuscular or functional* or movement*or walk* or gait* or balance or strength or tendon*or joint* or reflex* or tonus) **N3** (pain* or disorder* or disturb* or discomfort* or dysfunction* or disruption* or impair* or symptom* or change* or abnormal* or mechanical or deficit* or limit* or loss or diminish* or reduc* or alter* or disabilit* or postural or weak* or fatigue) or propriocepti* or arthralgia* or myalgia* or polyarthralgia* or numbness or foot drop or footdrop or cramp*) | |
| **26** | **OR/14-25** | **487721** |

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| | | |
|---|---|---|
| **CIPN** | | |
| **27** | **TI**((chemotherapy induced N2 neuropath*) or CIPN) | |
| **28** | **AB**((chemotherapy induced N2 neuropath*) or CIPN) | |
| **29** | **27 OR 28** | **780** |
| **Antineoplastic agents** | | |
| **30** | **MH**(antineoplastic agents+) | |
| **31** | **TI**(antineoplastic* or chemotherap* or ((anticancer or antitumo?) N3 (agent or agents or drug*))) | |
| **32** | **AB**(antineoplastic* or chemotherap* or ((anticancer or antitumo?) N3 (agent or agents or drug*))) | |
| **33** | **30 OR 31 OR 32** | **170559** |
| **34** | **4 AND 13 AND 26 AND 33** | **828** |
| **35** | **4 AND 13 AND 29** | **119** |
| **36** | **34 OR 35** | **829** |
| **Exclusion Filters** | | |
| **37** | **PT**(biography OR case study OR commentary OR directories OR editorial OR interview OR legal case OR letter OR practice guidelines) | |
| **38** | **MH**("Biographies+" OR "Legislation" OR "News+" OR "Serial Publications+") | |
| **39** | **37 or 38** | **1375622** |
| **40** | **36 not 39** | **806** |

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**PsycInfo (OVID)**

| | | |
|---|---|---|
| **Psychometrics properties** | | |
| **1** | exp psychometrics/ OR Interrater Reliability/ OR Discriminant Validity/ | **Results** |
| **2** | (reproducib* OR psychometr* OR clinimetr* OR clinometr* OR observer variation OR reliab* OR valid* OR coefficient OR internal consistency OR (cronbach* AND (alpha OR alphas)) OR item correlation* OR item selection* OR item reduction*).**ti,ab** OR (agreement OR precision OR imprecision OR precise values)**.tw** OR (test retest OR (test AND retest) OR (reliab* AND (test OR retest)) OR stability OR interrater OR inter rater OR intrarater OR intra rater OR intertester OR inter tester OR intratester OR intra tester OR interobserver OR inter observer OR intraobserver OR intra observer OR intertechnician OR inter technician OR intratechnician OR intra technician OR interexaminer OR inter examiner OR intraexaminer OR intra examiner OR interassay OR inter assay OR intraassay OR intra assay OR interindividual OR inter individual OR intraindividual OR intra individual OR interparticipant OR inter participant OR intraparticipant OR intra participant OR kappa OR kappa's OR kappas OR coefficient of variation).**ti,ab** OR (repeatab* OR ((replicab* OR repeated) AND (measure OR measures OR findings OR result OR results OR test OR tests))).**tw** OR (generaliza* OR generalisa* OR concordance OR (intraclass AND correlation*) OR discriminative OR known group OR factor analysis OR factor analyses OR factor structure OR factor structures OR dimensionality OR subscale* OR multitraitscaling analysis OR multitrait scaling analyses OR item discriminant OR interscale correlation* OR ((error OR errors) AND (measure*OR correlat* OR evaluat* OR accuracy OR accurate OR precision OR mean)) OR individual variability OR interval variability OR rate variability OR variability analysis OR (uncertainty AND (measurement OR measuring)) OR standard error of measurement OR sensitiv* OR responsive* OR (limit AND detection) OR minimal detectable concentration OR interpretab* OR (small* AND (real OR detectable) AND (change OR difference)) OR meaningful change OR minimal important change OR minimal important difference OR minimally important change OR minimally important difference OR minimal detectable change OR minimal detectable difference OR minimally detectable change OR minimally detectable difference OR minimal real change OR minimal real difference OR minimally real change OR minimally real difference OR ceiling effect OR floor effect OR Item response model OR IRT OR Rasch OR Differential item functioning OR DIF OR computer adaptive testing OR itembank OR cross cultural equivalence).**ti,ab** | |
| **3** | **1 OR 2** | **878016** |
| **Patient-Reported Outcome Measurement instruments (PROM)** | | |
| **4** | "Patient Reported Outcome Measures"/ or Surveys/ OR Questionnaires/ OR Self-Report/ OR exp Quality of Life/ OR Quality of Life Measures/ | |

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| **5** | ((patient? OR self) adj2 (report* OR relate* OR declar* OR rating* OR assess* OR examin* OR monitor* OR administer* OR rate* OR rating* or based or appraisal* or appraised or evaluat*) adj2 (outcome* OR symptom*) or PROM? Or PRO or PROS OR (quality adj2 life) OR HRQL or HRQoL or QL or QoL OR HR-PRO or HRPRO).**ti,ab** | |
| **6** | ((disability or function* or subjective or utility or utilities or wellbeing or well being) adj3 (outcome or outcomes or index or indices or instrument or instruments or measure or measures or questionnaire or questionnaires or profile or profiles or scale or scales or score or scores or scoring or status or survey or surveys or measure*)).**ti,ab** | |
| **Clinician reported outcomes measur** | | |
| **7** | Health Outcomes/ | |
| **8** | (Clinician* reported outcome* or CROS or clinRO or clinROs OR ((clinical or clinician* or expert* or health professional* OR physician* OR doctor*) adj2 (judgement* or interpretation* or observation* or assessment* or measure*))).**ti,ab** | |
| **9** | **OR/4-8** | **252244** |
| **Peripheral neuropathy** | | |
| **10** | Peripheral Neuropathy/ OR Neuropathic Pain/ OR Neuralgia/ OR "Complex Regional Pain Syndrome (Type I)"/ OR Central Nervous System Disorders/ | |
| **11** | (PNS disease* or peripheral nervous system disease* or neuropath* or complex regional pain syndrome* or neuralgia* or neuritis or neurodinya* or (nerve* adj2 (pain* or injur* or disease* or affect* or damage* or degeneration*)) or causalgia* or hyperesthesi* or paresthesi* or hypesthesi* or paralysis or polyneuropath* or polyradiculo* or polyneuritis or burning feet syndrome or metatarsalgia).**ti,ab** | |
| **12** | Nervous System Disorders/ | |
| **13** | ((neurologic* adj2 (manifestation* or deficit* or sign or signs or symptom* or finding* or dysfunction*))).**ti,ab** | |
| **14** | Somatosensory Disorders/ OR Pressure Sensation/ OR Pain Perception/ OR "Sensory Disabilities (Attitudes Toward)"/ OR Sensory Integration Dysfunction/ OR Sensory Gating/ OR Perceptual Disturbances/ OR Sensory System Disorders/ OR Pain Thresholds/ OR Thresholds/ OR Somatosensory Evoked Potentials/ OR "Signal Detection (Perception)"/ OR Evoked Potentials/ OR Somesthetic Perception/ OR Tactual Perception/ OR Somatoform Pain Disorder/ OR Somatoform Disorders/ OR Neurotoxicity/ | |
| **15** | (((sensation* or sense* or sensivity or sensorimotor or sensory or somatosensory or nociceptive or somatic or tissue* or tactile or touch* or pinprick or thermal or burning or shooting or deafferentation or differential or referred or percept* or facial or breakthrought) adj3 (pain* or disorder* or disease* or disturb*or distortion* or discomfort* or dysfunction* or disruption* or impair* or symptom* or change* or abnormal* or mechanical or deficit* or limit* or loss or diminish* or reduc* or amplificat* or hyperesthetic or alter* or affect* or gating or | |

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| | threshold*)) or pain threshold* or hyperalgesi* or allodynia* or formication* or tingling or hypersensitivity or oxyesthesi* or electric shock like or vibration* or nocicept* or neurotoxi* or hand foot syndrome*).**ti,ab** | |
| **16** | Musculoskeletal Disorders/ OR Neuromuscular Disorders/ OR Muscular Disorders/ OR Myasthenia/ OR Articulation Disorders/ OR Movement Disorders/ OR Proprioception/ | |
| **17** | ((motor* or muscle* or muscular or neuromuscular or functional* or movement*or walk* or gait* or balance or strength or tendon*or joint* or reflex* or tonus) adj3 (pain* or disorder* or disturb* or discomfort* or dysfunction* or disruption* or impair* or symptom* or change* or abnormal* or mechanical or deficit* or limit* or loss or diminish* or reduc* or alter* or disabilit* or postural or weak* or fatigue) or propriocepti* or arthralgia* or myalgia* or polyarthralgia* or numbness or foot drop or footdrop or cramp*).**ti,ab**. | |
| **18** | **OR/10-17** | **292620** |
| **CIPN** | | |
| **19** | ((chemotherapy induced adj2 neuropath*) or CIPN).**ti,ab** | |
| **Antineoplastic agents** | | |
| **20** | Antineoplastic Drugs/ OR Chemotherapy/ | |
| **21** | (antineoplastic* or chemotherap* or ((anticancer or antitumo?) adj3 (agent or agents or drug*))).**ti,ab** | |
| **22** | **20 OR 21** | **7227** |
| **23** | **3 AND 9 AND 18 AND 22** | **92** |
| **24** | **3 AND 9 AND 19** | **27** |
| **25** | **23 OR 24** | **92** |
| **Exclusion Filters** | | |
| **26** | Exp Biography/ OR Case Report/ OR exp laws/ OR Interviews/ OR Newspapers/ OR Treatment Guidelines/ OR Popular Culture/ | |
| **27** | **25 not 26** | **91** |

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BMJ Open

# Evaluation of the psychometric properties of patient-reported and clinician-reported outcome measures of chemotherapy-induced peripheral neuropathy: a COSMIN systematic review protocol

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| Primary Subject Heading: | Oncology |
| Secondary Subject Heading: | Palliative care |
| Keywords: | CHEMOTHERAPY, Cancer pain < ONCOLOGY, Adult oncology < ONCOLOGY, PAIN MANAGEMENT, Adult palliative care < PALLIATIVE CARE |
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# Evaluation of the psychometric properties of patient-reported and clinician-reported outcome measures of chemotherapy-induced peripheral neuropathy: a COSMIN systematic review protocol

Authors:

Philippe Bérubé-Mercier [1,2], Diane Tapp[1,3], Marie-Ève Cimon [1,3,] Tiffany Li [4], Susanna B. Park [4], Ève Bouhêlier [5], Kaitlin McGarragle [2], Lye-Ann Robichaud [6], Jennifer Gewandter [7], Maxime Bouchard, Lynn R. Gauthier [2,8]

1. Université Laval, Faculty of Nursing, Quebec, QC, Canada
2. Centre de recherche du CHU de Québec-Université Laval, Oncology Division, Quebec, QC, Canada
3. Quebec Heart and Lung Institute, Quebec, QC, Canada
4. The University of Sydney, School of Medical Sciences, Brain and Mind Centre, Sydney, NSW, Australia
5. CHU de Québec-Université Laval, Quebec, QC, Canada
6. Université de Montréal, Department of Psychology, Montreal, QC, Canada.
7. University of Rochester Medical Center, School of Medicine and Dentistry, Rochester, NY, United States
8. Université Laval, Department of Family and Emergency Medicine, Quebec City, Quebec, Canada

Contact Author : Lynn R. Gauthier
Hôtel-Dieu de Québec - CRCEO
6, rue McMahon, local 1899-5
Quebec City, Québec
Canada G1R 3S1
Email : Lynn.Gauthier@crchudequebec.ulaval.ca

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## Abstract

**Introduction:** Chemotherapy-induced peripheral neuropathy (CIPN) is a poorly understood side effect of many antineoplastic agents. Patients may experience sensory, motor, and autonomic symptoms, negatively impacting quality of life. A gold-standard assessment methodology has yet to be determined, limiting efforts to identify effective agents to prevent or treat CIPN.

**Methods and Analysis:** This is a protocol of a systematic review of psychometric analyses of CIPN Clinician Reported Outcome Measures (ClinROM) and Patient-Reported Outcome Measures (PROM) among adults receiving, or who had previously received chemotherapy for cancer. The COnsensus-based Standards for the selection of health Measurement INstruments (COSMIN) quality ratings will be compared across studies and across ClinROMs and PROMs. Studies reporting psychometric proprieties of CIPN ClinROMs and/or PROMs among adults aged ≥ 18 years will be eligible for inclusion, with no restriction on language or year of publication. MEDLINE, Embase, CINAHL, and APA PsycInfo databases will be searched from inception to December 31st, 2021. Study characteristics, measurement properties of the ClinROMs and/or PROMs and the CIPN definitions will be extracted. The Synthesis Without Meta-analysis guideline will be used to guide data synthesis. The COSMIN Risk of Bias checklist will be used by two independent raters to assess methodological quality. Subgroup analyses by age, chemotherapy type, and study timing in relation to the delivery of chemotherapy will be carried out where data are available. An adapted version of Outcome Measures in Rheumatology (OMERACT) filter 2.1 will be used to provide a best-evidence synthesis of CIPN ClinROMs and PROMs and to recommend a CIPN assessment tool for clinical and research settings.

**Ethics and Dissemination:** Ethical approval is not necessary to be obtained for this systematic review protocol. Results will be disseminated to clinicians and policymakers by publication in a peer-reviewed journal and by presenting at relevant conferences.

**Systematic Review Registration:** PROSPERO number CRD42021278168

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**Strengths and Limitations**

- This proposed study will be the most-up-to date systematic review of CIPN ClinROMs and PROMs and the first to use COSMIN methodology to assess the methodological quality of studies reporting psychometric properties of CIPN ClinROMs.
- The proposed study is also the first to use the OMERACT filter 2.1 methodology following COSMIN analysis of CIPN ClinROMs and PROMs facilitating the recommendation of one or more ClinROMs and/or PROMs for assessing CIPN, based on the available evidence.
- Subgroup analyses by age group, chemotherapy type, and the timing of the study in relation to the delivery of chemotherapy will be undertaken to gain a better understanding of potential subgroup differences in the psychometric proprieties of CIPN ClinROMs and PROMs.
- Limitations may be related to the heterogeneity of patients who have CIPN and to the lack of consensus about the definition of CIPN.
- As the COSMIN methodology does not consider patient-reported experience measures (PREMs)**,** they are not included in this review, potentially impacting the identification of a tool that could improve patient health outcomes.

**Keywords**

COSMIN, Psychometric analysis, chemotherapy-induced peripheral neuropathy, patient-reported outcome measures, clinician-reported outcome measures

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## Introduction

Chemotherapy-induced peripheral neuropathy (CIPN) is a challenging and common side effect of many antineoplastic agents with significant negative impacts on quality of life (1-3). Patients may experience an array of symptoms such as numbness, tingling, pins-and-needles, burning, motor weakness, and/or balance disturbance (1). A meta-analysis has suggested that 68.1% (95% Confidence Interval [CI]: 57.7-78.4) of patients have CIPN symptoms in the first month after completing chemotherapy and 60.0% (95% CI: 36.4-81.6) of patients have symptoms three months after finishing chemotherapy (3, 4). Thirty percent (95% CI: 6.4-53.5) of patients report persistent CIPN symptoms 6 months or later after treatment (4). Wide confidence intervals suggest substantial uncertainty in this estimate, potentially due to variability in measurement tools used in these studies (5, 6). CIPN may lead to dose reductions, delayed treatment or early discontinuation in 2% to 58% of people, with unclear effects on disease status and survival (7-13). Unfortunately, there are no known preventative agents, and only one pharmacologic treatment, duloxetine, with sufficient evidence supporting its use when the outcome is pain intensity (14, 15). However, treatments for the myriad other symptoms associated with CIPN are unavailable.

Early, frequent, and standardized assessment of CIPN is important to understand the development and evolution of symptomatology and to prevent its negative impacts (3, 14). This may be achieved via use of Clinician-Reported Outcome Measures (ClinROM) or Patient-Reported Outcome Measures (PROM). ClinROMs are assessment tools and techniques used by clinicians of observable signs, behaviors, or other manifestations of CIPN (16-20). PROMs allow patients to self-report the presence, intensity, frequency, and/or the impact of symptoms and signs of the sensory, motor, and/or autonomic features of CIPN. CIPN ClinROMs and PROMs must be valid, or measure what they purport to measure(21), reliable (i.e., consistent), reproducible, and stable across raters (22), and responsive to change following administration of the first and subsequent cycles of neurotoxic chemotherapy (20). Unfortunately, there is no agreement on a gold standard CIPN assessment tool, preventing an adequate understanding of the magnitude of the problem of acute and chronic CIPN, its presentation, development and trajectory, risk

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factors, and impacts (20, 23). Importantly, this may also contribute to difficulty identifying effective preventative agents and treatments (14, 23, 24).

It is also unclear whether CIPN tools have similar validity and reliability across neurotoxic chemotherapy types, during versus after chemotherapy delivery, or across the adult lifespan (3). For example, while there may be heterogeneity in CIPN experience due to biopsychosocial factors (e.g. genetics, symptom appraisal) (4, 8), CIPN may also differ according to chemotherapy type with potentially different effects on symptom presentation (3). It may also be possible that the psychometric properties of the tools may differ depending on whether CIPN is measured during treatment (e.g. acute symptom presentation) versus after treatment (e.g. chronic symptom presentation). Additionally, aging is accompanied by non-uniform biopsychosocial changes which can influence health and illness (25). In the case of CIPN, age-related morphological changes in the peripheral nervous system, including decreased nerve conduction velocity and slowed axonal regeneration and reinnervation (26-31), may interact with damage to the peripheral nervous system caused by neurotoxic chemotherapy (3), which may also result in differences in symptom presentation. This is important because cancer is primarily a disease of older people (32, 33). Therefore, it may be possible that in the upcoming years, with the aging population (34), there will be many older adults at risk of experiencing CIPN who will require valid and reliable assessment strategies that are able to differentiate normal age-related changes from those associated with CIPN (32, 35).

Although numerous reviews of CIPN ClinROMs and PROMs have been published (6, 36-40), none to date has used methodology that allows for a quantitative assessment and comparison of the methodological quality of psychometric studies or the quality of the CIPN assessment tools, themselves. This is likely a key factor in our inability to identify a gold standard CIPN tool. The COnsensus-based Standards for the selection of health Measurement INstruments (COSMIN) Risk of Bias checklist was developed to quantitatively assess the methodological quality of studies reporting psychometric properties and the ClinROMs and PROMs evaluated in these studies (16, 41, 42). Validity, reliability, and responsiveness are evaluated using 10 boxes that collect information about

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measurement properties, including measure development, content validity, structural validity, internal consistency, cross-cultural validity/measurement invariance, reliability, measurement error, criterion validity, hypothesis testing, and responsiveness. Assessment criteria in each box depend on the type of analysis performed. The scoring of each question within a box can vary according the measurement propriety assessed (41). The rating scale includes four possible scores, ranging from very good, adequate, doubtful, to inadequate (41). The score for each box is calculated based on a "worst score counts" method which only considers the lowest rating within the box. This rigorous method allows for an objective evaluation and quantitative scoring of the quality of studies that report psychometric properties of CIPN ClinROMs and PROMs and the tools themselves. No studies to date have applied the COSMIN methodology to assess the methodological quality of studies reporting psychometric properties of CIPN tools and the overall quality of ClinROMs and/or PROMs.

There are significant gaps in our knowledge about how to best assess CIPN. These knowledge gaps affect our capacity to relieve CIPN symptoms in all adults undergoing neurotoxic chemotherapy. A critical first step to improving our understanding of CIPN across the adult lifespan is to ensure that the assessment tools are valid and reliable. An assessment of the methodological quality of studies reporting psychometric properties of CIPN tools and the quality of ClinROMs and PROMs using the COSMIN method would help to identify gaps and weaknesses in their psychometric properties and potential areas for improvements to existing CIPN measures. Such an analysis would substantially improve the CIPN literature. From a research perspective, it would advance the field toward identification of a gold standard CIPN assessment method, which could have important implications for future trials testing novel preventive and treatment agents. From a clinical perspective, it would provide healthcare providers with valid and reliable assessment tools for early detection and monitoring of CIPN symptoms among younger and older adults undergoing all neurotoxic chemotherapies.

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Objectives

The primary objective of this study is to perform a systematic review of studies reporting psychometric properties of CIPN ClinROMs and PROMs among adults receiving, or who had previously received chemotherapy for cancer and to compare COSMIN quality ratings across included studies and the quality of ClinROMs and PROMs, separately. The secondary objective is to perform subgroup analyses of studies by age group (younger adults [18 years old - 59 years old ± 10 years] vs. older adults [≥60 years old ±10 years]), chemotherapy type, and the timing of the study in relation to the delivery of chemotherapy (e.g. during treatment vs after treatment), where data are available.

**Methods and Analysis**

This study protocol was prepared with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols (PRISMA-P) and Preferred Reporting Items for Systematic reviews and Meta-Analyses literature search extension (PRISMA-S) guidelines (43, 44). The PRISMA-P checklist is available in Appendix 1. This systematic review protocol was registered with the International Prospective Register of Systematic Reviews (PROSPERO) on September 13, 2021. Any changes of this protocol will be documented and recorded in PROSPERO. The start date is July 9, 2021, and the anticipated completion date is April 2022.

PICO Question

- Population: Adults aged over 18 years receiving or who had previously received chemotherapy for cancer
- Intervention(s), exposure(s): Any type of PROM or ClinROM used to assess CIPN
- Comparator(s)/control: Not applicable
- Main outcome(s): Establishing what is the most appropriate, valid and reliable CIPN PROM and ClinROM using the COSMIN methodology
- Additional outcome: Recommend of one or more ClinROM(s) and/or a PROM(s) for assessing CIPN in clinical and research settings using the OMERACT filter 2.1 methodology.

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Eligibility Criteria

All studies reporting on the psychometric proprieties of CIPN ClinROMs and/or PROMs among adults aged ≥ 18 years will be eligible for inclusion. There will be no restrictions on publication or year. Articles published in French will be translated to English by bilingual team members (PMB, LRG, DT, MEC) (45). For articles written in other languages we will use Google Translate, as recommended (45). All types of cancer and all types of chemotherapy will be considered. Only empirical, peer-reviewed articles that report on measurement proprieties of CIPN ClinROMs and/or PROMs will be included. Case studies, study protocols, published conference abstracts and systematic reviews will be excluded. Studies reporting data from neonatal and/or pediatric populations combined with data from adult populations will be included if it is possible to isolate the data from the adult subgroup. If it is not possible, these studies will be excluded. Studies will be excluded when an adult proxy is used to assess CIPN symptoms of pediatric patients. Patients who do not have cancer or who do not receive any antineoplastic agents will be excluded. Any studies using animals as their population will also be excluded

Information Sources

MEDLINE, Embase, CINAHL, and APA PsycInfo databases will be searched from inception to December 31st, 2021. Search terms were developed for 1) the population (adults receiving or who had previously received cancer treatment), 2) CIPN, and 3) psychometric properties. The COSMIN search filter for PROMs and ClinROMs was used for the latter conceptual block. Search strategies were verified by a health science librarian with systematic review experience. References of all identified texts will be examined to identify additional studies not identified in the database search.

Search Strategy: Search strategies are available in Appendix 2.

Study Records: PROSPERO number CRD42021278168

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Data Management

All references will be imported into Endnote v20. Duplicates will be removed using Bramer's de-duplication method in Endnote (46). Then, references will be exported into Covidence in order to carry out the study selection (47).

Study Selection Process

A two-stage process for study screening will be used. Two independent raters (PBM, LAR) will screen titles and abstracts of articles to assess whether they meet the eligibility criteria. Full-text screening will be conducted by the 2 independent raters (PBM, KM). The number of included and excluded studies at each step will be presented in a PRISMA flow chart. Reasons for the exclusion of full-text articles will be recorded in Covidence and presented in the PRISMA flow chart. A third rater (LRG) will resolve all disagreements at each screening phase. Percentage agreement between raters will be calculated at each phase to report interrater reliability (48).

Data Collection Process

For each study included in the review, data relating to study characteristics (target population, mode of administration, recall period, subscale(s) and number of items, response options, range of scores, original language and available translation) and measurement properties of the PROMs and the ClinROMs (instrument development, content validity, structural validity, internal consistency, cross-cultural validity/measurement invariance, reliability, measurement error, criterion validity, hypothesis testing for the construct validity and responsiveness) will be extracted by 2 people and compared for consistency (PBM, MEC). CIPN definitions reported by the authors will also be extracted to examine and compare conceptual definitions across studies. Data extraction tables will be adapted from those designed by the COSMIN group. Corresponding authors will be contacted in the event of missing data (descriptive data, missing information about the items and the scoring of each tool). Three contact attempts will be made over the period of one month. In the case of non-response, data will be coded as missing.

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Quality Assessment

The COSMIN Risk of Bias checklist will be used by two independent raters (PBM, MEC) to assess the methodological quality of the included studies and the quality of ClinROMs and PROMs (42). The first step consists of assessing the quality of each study describing psychometric properties of CIPN ClinROMs and PROMs using an Excel spreadsheet proposed by the COSMIN group (49). In this step, each of the 10 measurement properties considered by the COSMIN system is given 4 possible scores: very good, adequate, doubtful and inadequate (49). A summary table which presents the quality rating of each study will be included in this review once the COSMIN Risk of Bias checklist is completed. The second step involves rating each study describing psychometric properties of CIPN ClinROMs and PROMs against the criteria for good measurement properties (49). Each rating has a possible score of sufficient (+), insufficient (-), inconsistent (±) or indeterminate rating (?) according to published criteria (49). The third step involves determining an overall quality score of all studies reporting on a given ClinROM or PROM by considering cross-study ratings (49). The quality of evidence is determined by using a modified GRADE approach with a possibility of 4 grades: high, moderate, low, very low (49). Psychometric data from ClinROMs will be considered separately from psychometric data from PROMs. We will rely on data for steps 1 and 2 from a recently completed systematic review and COSMIN analysis of CIPN PROMs (PROSPERO 2020 CRD42020210405; Li, Park & Rutherford, Manuscript in progress) for any studies overlapping across the 2 reviews.

Subgroup Analysis

The COSMIN Risk of Bias checklist will be completed and analyzed to compare methodological quality ratings from studies reporting psychometric properties of CIPN tools and the quality of ClinROMs and PROMs across age groups (younger adults [18 years old - 59 years old ± 10 years] vs. older adults [≥60 years old ±10 years]), the type of chemotherapy received, and the timing of the study in relation to the delivery of chemotherapy (e.g. during treatment vs after treatment), where data are available. Age subgroups are based on cut-offs used in previous studies in cancer with clinically relevant outcomes (50-52). We will allow for the cutoff between younger and older groups to vary

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by 20 years to account for methodological differences in the categorization of younger and older age groups across studies.

Recommendation of a ClinROM and PROM

While the COSMIN method is the only available method to quantitatively assess the methodological quality of studies reporting psychometric properties and the ClinROMs and PROMs, it does not provide a method to recommend the use of a ClinROMs or PROMs. In this review, we will address this limitation by using the Outcome Measures in Rheumatology (OMERACT) filter 2.1, based on the findings of the COSMIN analysis, to determine whether it is possible to recommend the use of one or more ClinROMs or PROMs, and if so, make recommendations (53, 54). Data issued by COSMIN on construct validity, test-retest reliability and longitudinal construct validity (i.e. responsiveness) and additional information on clinical trial discrimination and thresholds of meaning which are not covered in COSMIN will be used to complete the COSMIN-OMERACT Good Measures Checklist by two independent raters (53). This methodology permits recommendations based on a standardized, rigorous and transparent process (53). Although this methodology has not been used in systematic reviews of the psychometric properties of CIPN measures, 2 previous reviews have used the OMERACT filter 2.1 in conjunction with the results of a COSMIN analysis to make recommendations about dermatological measures (55, 56).

The OMERACT filter 2.1 (Table 1) has 3 pillars, 4 questions, 7 measurement properties and 1 answer (53). Truth, discrimination, and feasibility represent the pillars (53). A set of questions are associated with each pillar which are organized in an algorithm (53). Each question has a possibility of 4 scores: Red, amber, green, and white (53). Red means “stop, do not continue”, amber means “a caution is raised but you can continue”, green means “go, this question is definitely answered affirmatively”, and white means an absence of evidence and the evaluation must stop (53). An overall score that ranges between do not endorse, provisionally endorse, or endorse is issued once the 4 questions of the algorithm are answered (53). The 7 included measurement properties map onto 7 of the COSMIN

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measurement properties. We will adapt the OMERACT filter 2.1 to include all 10 COSMIN Risk of bias checklist measurement properties.

**Table 1 OMERACT filter 2.1 definitions, questions, and measurement properties**

| **Pillars** | **Definition** | **Questions** | **Measurement properties** |
|---|---|---|---|
| Truth | The ability of the outcome measurement tool to measure what is intended (53) | "Is it a match with the target domain?" | Construct validity, reliability |
| | | "Do the numeric scores make sense?" | Content validity, face validity |
| Discrimination | The ability of the outcome measurement tool to discriminate different situations of interest (53) | "Can it discriminate between groups of interest?" | Test-retest reliability, longitudinal validity/responsiveness, ability to discriminate in RCT/comparative research setting, threshold of meaning |
| Feasibility | The practicality of the outcome measurement tool (time, cost, burden) (53) | "Is it practical to use?" | Access, training, translation, length, cost, burden |

Data Synthesis and Best-Evidence Synthesis

The Synthesis Without Meta-analysis (SWiM) guideline will be used to guide data synthesis for the included studies (57). SWiM includes 9 items that report key features in the methods, results and discussion of every study, such as grouping studies for synthesis (item 1), standardized metric used for synthesis (item 2), synthesis methods and their limitations (items 3 and 9), criteria used to prioritize results for summary and synthesis

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(item 4), heterogeneity in reported effects (item 5), certainty of evidence (item 6), data presentation methods (item 7) and a summary of the synthesis (item 8) (57).

Patient and Public Involvement

A patient partner and co-author (MB) participated in the development of the review protocol and will participate as a research team member throughout the review process, contributing to interpretation of the findings, manuscript drafting and revisions.

**Ethics and dissemination**

An ethical approval is not necessary to be obtained for this systematic review protocol. The results will be disseminated to clinicians and researchers by publication in a peer-reviewed journal and by presenting at relevant conferences. This systematic review will support researchers and clinicians to use the best measure to assess CIPN.

**Discussion**

The planned study will be the most-up-to date systematic review of CIPN ClinROMs and PROMs and the only one to use the COSMIN risk of bias tool with CIPN ClinROMs, allowing for a quantitative evaluation and comparison of the methodological quality of studies reporting psychometric properties of CIPN tools and the quality of ClinROMs and PROMs. Subgroup analyses by age group, chemotherapy type, and the timing of the study in relation to the delivery of chemotherapy will be carried out to determine whether the psychometric proprieties of CIPN ClinROMs and PROMs are the same across clinically relevant subgroups. This systematic review will also be the first to use the OMERACT filter 2.1 methodology to facilitate the recommendation of one or more ClinROMs and/or PROMs for assessing CIPN in clinical and research settings, based on the available evidence.

Limitations may be related to the heterogeneity of patients who have CIPN and to the lack of consensus about the definition of CIPN. High heterogeneity in the experience of CIPN may be due to multiple factors, (e.g. biological, the prescribed chemotherapy regimen, and the timing of development of CIPN and its manifestations over time. Our planned subgroup

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analyses may help to mitigate this limitation. We will also analyze CIPN definitions used across studies to examine conceptual definitions, and to compare similarities and differences across studies and how this might contribute to any observed heterogeneity. In addition, the COSMIN methodology does not currently consider all types of measurement like patient-reported experience measures (PREM). PREMs aim to explore the patient's experience of care from their own perspective (58). This limitation may have an impact on the identification of a tool that could improve patient health outcomes and the quality of care for those undergoing chemotherapy or receiving CIPN treatment and the economic impact of CIPN (58, 59). Additional research is needed to identify and evaluate the quality of CIPN PREMs. Expanding the COSMIN methodology to include PREMs would be an important future research direction.

## Declarations

### Competing interests

The authors declare that they have no competing interests.

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### Contributors

PBM and LRG conceived the study idea and were responsible for developing and writing the first draft of the systematic review protocol and manuscript. EB and PBM developed

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search routine for each database. TL, SBP, LAR, MEC, KM, DT, MB and JG provided critical insights at all stages. All authors approved and contributed to the final manuscript.

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**Appendix 1: PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol***

| Section and topic | Item No | Checklist item | Page number |
|---|---|---|---|
| **ADMINISTRATIVE INFORMATION** | | | |
| Title: | | | |
| Identification | 1a | Identify the report as a protocol of a systematic review | Title; p.1 |
| Update | 1b | If the protocol is for an update of a previous systematic review, identify as such | n/a |
| Registration | 2 | If registered, provide the name of the registry (such as PROSPERO) and registration number | Abstract, Methods and Analysis, Study Records; pp.2,7,8 |
| Authors: | | | |
| Contact | 3a | Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author | Authors; p.1 |
| Contributions | 3b | Describe contributions of protocol authors and identify the guarantor of the review | Declaration; Authors' Contributions: p.13 |
| Amendments | 4 | If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments | n/a |
| Support: | | | |
| Sources | 5a | Indicate sources of financial or other support for the review | Declaration; Funding Statement; p.12 |
| Sponsor | 5b | Provide name for the review funder and/or sponsor | Declaration; Funding |

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|---|---|---|---|
| | | | Statement; p.12 |
| Role of sponsor or funder | 5c | Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol | Declaration; Funding statement; p.12 |
| **INTRODUCTION** | | | |
| Rationale | 6 | Describe the rationale for the review in the context of what is already known | Introduction; pp.4-6 |
| Objectives | 7 | Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO) | Methods and Analysis; PICO Question; p.7 |
| **METHODS** | | | |
| Eligibility criteria | 8 | Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review | Methods and Analysis; pp.7-8 |
| Information sources | 9 | Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage | Methods and Analysis; Information Sources; pp.7-8 |
| Search strategy | 10 | Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated | Supplementary material file; pp.5-18 |
| Study records: | | | |
| Data management | 11a | Describe the mechanism(s) that will be used to manage records and data throughout the review | Methods and Analysis; pp.7-9 |
| Selection process | 11b | State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis) | Methods and Analysis; Study |

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| | | | Selection Process; p.8 |
| Data collection process | 11c | Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators | Methods and Analysis; Data Collection Process; p.9 |
| Data items | 12 | List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications | pp.6, 8 |
| Outcomes and prioritization | 13 | List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale | Methods and Analysis; PICO Question; p.6 |
| Risk of bias in individual studies | 14 | Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis | Methods and Analysis; Quality Assessment; pp.9-10 |
| Data synthesis | 15a | Describe criteria under which study data will be quantitatively synthesised | Methods and analysis; Data Synthesis and Best-Evidence Synthesis; p.11 |
| | 15b | If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as $I^2$, Kendall's $\tau$) | Methods and Analysis; Data Synthesis and Best-Evidence Synthesis; p.11 |
| | 15c | Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression) | Methods and Analysis; Subgroup Analysis; p.10 |
| | 15d | If quantitative synthesis is not appropriate, describe the type of summary planned | Methods and Analysis; |

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|---|---|---|---|
| | | | Quality Assessment; pp.9-10 |
| Meta-bias(es) | 16 | Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies) | n/a |
| Confidence in cumulative evidence | 17 | Describe how the strength of the body of evidence will be assessed (such as GRADE) | n/a |

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

*From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647*

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## Appendix 2: Search strategies

### Medline (OVID)

| Psychometrics properties | | Results |
|---|---|---|
| **1** | instrumentation.fs OR validation study.pt OR exp reproducibility of results/ OR psychometrics/ OR observer variation/ OR discriminant_analysis/ | |
| **2** | (reproducib* OR psychometr* OR clinimetr* OR clinometr* OR observer variation OR reliab* OR valid* OR coefficient OR internal consistency OR (cronbach* AND (alpha OR alphas)) OR item correlation* OR item selection* OR item reduction*).**ti,ab** OR (agreement OR precision OR imprecision OR precise values).**tw** OR (test retest OR (test AND retest) OR (reliab* AND (test OR retest)) OR stability OR interrater OR inter rater OR intrarater OR intra rater OR intertester OR inter tester OR intratester OR intra tester OR interobserver OR inter observer OR intraobserver OR intra observer OR intertechnician OR inter technician OR intratechnician OR intra technician OR interexaminer OR inter examiner OR intraexaminer OR intra examiner OR interassay OR inter assay OR intraassay OR intra assay OR interindividual OR inter individual OR intraindividual OR intra individual OR interparticipant OR inter participant OR intraparticipant OR intra participant OR kappa OR kappa's OR kappas OR coefficient of variation).**ti,ab**<br>OR (repeatab* OR ((replicab* OR repeated) AND (measure OR measures OR findings OR result OR results OR test OR tests))).**tw**<br>OR (generaliza* OR generalisa* OR concordance OR (intraclass AND correlation*) OR discriminative OR known group OR factor analysis OR factor analyses OR factor structure OR factor structures OR dimensionality OR subscale* OR multitraitscaling analysis OR multitrait scaling analyses OR item discriminant OR interscale correlation* OR ((error OR errors) AND (measure*OR correlat* OR evaluat* OR accuracy OR accurate OR precision OR mean)) OR individual variability OR interval variability OR rate variability OR variability analysis OR (uncertainty AND (measurement OR measuring)) OR standard error of measurement OR sensitiv* OR responsive* OR (limit AND detection) OR minimal detectable concentration OR interpretab* OR (small* AND (real OR detectable) AND (change OR difference)) OR meaningful change OR minimal important change OR minimal important difference OR minimally important change OR minimally important difference OR minimal detectable change OR minimal detectable difference OR minimally detectable change OR minimally detectable difference OR minimal real change OR minimal real difference OR minimally real change OR minimally real difference OR ceiling effect OR floor effect OR Item response model OR IRT OR Rasch OR Differential item functioning OR DIF OR computer adaptive testing OR itembank OR cross cultural equivalence).**ti,ab** | |
| **3** | **1 OR 2** | **4274030** |

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| | | |
|---|---|---|
| **Patient-reported outcome measurement instruments** | | |
| **4** | "patient reported outcome measures"/ or "surveys and questionnaires"/ or self report/ or quality of life/ | |
| **5** | (((patient? OR self) **adj2** (report* OR relate* OR declar* OR rating* OR assess* OR examin* OR monitor* OR administer* OR rate* OR rating* or based or appraisal* or appraised or evaluat*) **adj2** (outcome* OR symptom*)) or PROM? Or PRO or PROS OR (quality adj2 life) OR HRQL or HRQoL or QL or QoL OR HR-PRO or HRPRO).**ti,ab,kw** | |
| **6** | ((disability or function* or subjective or utility or utilities or wellbeing or well being) **adj3** (outcome or outcomes or index or indices or instrument or instruments or measure or measures or questionnaire or questionnaires or profile or profiles or scale or scales or score or scores or scoring or status or survey or surveys or measure*)).**ti,ab,kw** | |
| **Clinician reported outcome measurement instruments** | | |
| **7** | exp outcome assessment health care/ | |
| **8** | (Clinician* reported outcome* or CROS or clinRO or clinROs OR ((clinical or clinician* or expert* or health professional* OR physician* OR doctor*) **adj2** (judgement* or interpretation* or observation* or assessment* or measure*))).**ti,ab** | |
| **9** | **OR/4-8** | **2423430** |
| **Peripheral neuropathy** | | |
| **10** | peripheral nervous system diseases/ or brachial plexus neuropathies/ or exp complex regional pain syndromes/ or neuralgia/ or metatarsalgia/ or neuritis/ or polyneuropathies/ or polyradiculoneuropathy/ or small fiber neuropathy/ or neural conduction/ or cranial nerve injuries/ or peripheral nerve injuries/ | |
| **11** | (PNS disease* or peripheral nervous system disease* or neuropath* or complex regional pain syndrome* or neuralgia* or neuritis or neurodinya* or (nerve* **adj2** (pain* or injur* or disease* or affect* or damage* or degeneration*)) or causalgia* or hyperesthesi* or paresthesi* or hypesthesi* or paralysis or polyneuropath* or polyradiculo* or polyneuritis or burning feet syndrome or metatarsalgia).**ti,ab** | |
| **12** | neurologic manifestations/ | |
| **13** | ((neurologic* **adj2** (manifestation* or deficit* or sign or signs or symptom* or finding* or dysfunction*))).**ti,ab** | |
| **14** | sensation disorders/ or exp somatosensory disorders/ or sensory thresholds/ or differential threshold/ or pain threshold/ or signal detection, psychological/ or "evoked potentials, somatosensory"/ or touch perception/ or exp nociceptive pain/ or pain, referred/ or exp pain perception/ or breakthrough pain/ or facial pain/ or neurotoxicity syndromes/ or hand-foot syndrome/ | |
| **15** | (((sensation* or sense* or sensivity or sensorimotor or sensory or somatosensory or nociceptive or somatic or tissue* or tactile or touch* or pinprick or thermal or burning or shooting or deafferentation or differential or referred or percept* or facial or breakthrought) **adj3** (pain* or disorder* or disease* or disturb*or distortion* | |

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|---|---|---|
| | or discomfort* or dysfunction* or disruption* or impair* or symptom* or change* or abnormal* or mechanical or deficit* or limit* or loss or diminish* or reduc* or amplificat* or hyperesthetic or alter* or affect* or gating or threshold*)) or pain threshold* or hyperalgesi* or allodynia* or formication* or tingling or hypersensitivity or oxyesthesi* or electric shock like or vibration* or nocicept* or neurotoxi* or hand foot syndrome*).**ti,ab** | |
| **16** | exp "musculoskeletal and neural physiological phenomena"/ or muscular diseases/ or neuromuscular manifestations/ or muscle weakness/ or muscle cramp/ or arthralgia/ or exp gait disorders, neurologic/ or proprioception/ or weight-bearing/ or evoked potentials, motor/ | |
| **17** | ((motor* or muscle* or muscular or neuromuscular or functional* or movement*or walk* or gait* or balance or strength or tendon*or joint* or reflex* or tonus) **adj3** (pain* or disorder* or disturb* or discomfort* or dysfunction* or disruption* or impair* or symptom* or change* or abnormal* or mechanical or deficit* or limit* or loss or diminish* or reduc* or alter* or disabilit* or postural or weak* or fatigue) or propriocepti* or arthralgia* or myalgia* or polyarthralgia* or numbness or foot drop or footdrop or cramp*)**.ti,ab.** | |
| **18** | **OR/10-17** | **2942852** |
| **CIPN** | | |
| **19** | ((chemotherapy induced adj2 neuropath*) or CIPN).**ti,ab** | |
| **Antineoplastic agents** | | |
| **20** | Exp antineoplastic agents/ or antineoplastic combined chemotherapy protocols/ | |
| **21** | (antineoplastic* or chemotherap* or ((anticancer or antitumo?) **adj3** (agent or agents or drug*))).**ti,ab** | |
| **22** | **20 OR 21** | **1408051** |
| **23** | **3 AND 9 AND 18 AND 22** | **2741** |
| **24** | **3 AND 9 AND 19** | **167** |
| **25** | **23 OR 24** | **2741** |
| **Exclusion Filters** | | |
| **26** | (address OR case reports OR comment OR directory OR editorial OR festschrift OR lectures OR legal cases OR legislation OR letter OR news OR newspaper article OR patient education handout OR popular work OR congress OR practice guideline).**pt** | |
| **27** | exp biography/ OR exp consensus development conference/ | |
| **28** | **26 or 27** | **4234403** |
| **29** | **25 NOT 28** | **2562** |

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**Embase (OVID)**

| Psychometrics properties | | Results |
|---|---|---|
| **1** | devices/ OR validation study/ OR reproducibility/ OR exp psychometry/ or observer variation/ OR discriminant analysis/ OR validity/ | |
| **2** | (reproducib* OR psychometr* OR clinimetr* OR clinometr* OR observer variation OR reliab* OR valid* OR coefficient OR internal consistency OR (cronbach* AND (alpha OR alphas)) OR item correlation* OR item selection* OR item reduction*).**ti,ab** OR (agreement OR precision OR imprecision OR precise values).**tw** OR (test retest OR (test AND retest) OR (reliab* AND (test OR retest)) OR stability OR interrater OR inter rater OR intrarater OR intra rater OR intertester OR inter tester OR intratester OR intra tester OR interobserver OR inter observer OR intraobserver OR intra observer OR intertechnician OR inter technician OR intratechnician OR intra technician OR interexaminer OR inter examiner OR intraexaminer OR intra examiner OR interassay OR inter assay OR intraassay OR intra assay OR interindividual OR inter individual OR intraindividual OR intra individual OR interparticipant OR inter participant OR intraparticipant OR intra participant OR kappa OR kappa's OR kappas OR coefficient of variation).**ti,ab**<br>OR (repeatab* OR ((replicab* OR repeated) AND (measure OR measures OR findings OR result OR results OR test OR tests))).**tw**<br>OR (generaliza* OR generalisa* OR concordance OR (intraclass AND correlation*) OR discriminative OR known group OR factor analysis OR factor analyses OR factor structure OR factor structures OR dimensionality OR subscale* OR multitraitscaling analysis OR multitrait scaling analyses OR item discriminant OR interscale correlation* OR ((error OR errors) AND (measure*OR correlat* OR evaluat* OR accuracy OR accurate OR precision OR mean)) OR individual variability OR interval variability OR rate variability OR variability analysis OR (uncertainty AND (measurement OR measuring)) OR standard error of measurement OR sensitiv* OR responsive* OR (limit AND detection) OR minimal detectable concentration OR interpretab* OR (small* AND (real OR detectable) AND (change OR difference)) OR meaningful change OR minimal important change OR minimal important difference OR minimally important change OR minimally important difference OR minimal detectable change OR minimal detectable difference OR minimally detectable change OR minimally detectable difference OR minimal real change OR minimal real difference OR minimally real change OR minimally real difference OR ceiling effect OR floor effect OR Item response model OR IRT OR Rasch OR Differential item functioning OR DIF OR computer adaptive testing OR itembank OR cross cultural equivalence).**ti,ab** | |
| **3** | **1 OR 2** | **5504013** |
| **Patient-Reported outcome measurement** | | |
| **4** | patient-reported outcome measure/ OR questionnaire/ OR self report/ OR quality of life/ | |

| | | |
|---|---|---|
| **5** | (((patient? OR self) **adj2** (report* OR relate* OR declar* OR rating* OR assess* OR examin* OR monitor* OR administer* OR rate* OR rating* or based or appraisal* or appraised or evaluat*) **adj2** (outcome* OR symptom*) or PROM? Or PRO or PROS OR (quality adj2 life) OR HRQL or HRQoL or QL or QoL OR HR-PRO or HRPRO).**ti,ab,kw** | |
| **6** | ((disability or function* or subjective or utility or utilities or wellbeing or well being) adj3 (outcome or outcomes or index or indices or instrument or instruments or measure or measures or questionnaire or questionnaires or profile or profiles or scale or scales or score or scores or scoring or status or survey or surveys or measure*)).**ti,ab,kw** | |
| **Clinician reported outcomes measurement** | | |
| **7** | exp outcome assessment/ | |
| **8** | (Clinician* reported outcome* or CROS or clinRO or clinROs OR ((clinical or clinician* or expert* or health professional* OR physician* OR doctor*) **adj2** (judgement* or interpretation* or observation* or assessment* or measure*))).**ti,ab** | |
| **9** | **OR/4-8** | **2786142** |
| **Peripheral neuropathy** | | |
| **10** | peripheral neuropathy/ OR brachial plexus neuropathy/ OR exp complex regional pain syndrome/ OR neuralgia/ OR metatarsalgia/ OR neuritis/ OR polyneuropathy/ OR polyradiculoneuropathy/ OR small fiber neuropathy/ OR nerve conduction/ OR cranial nerve injury/ OR peripheral nerve injury/ | |
| **11** | (PNS disease* or peripheral nervous system disease* or neuropath* or complex regional pain syndrome* or neuralgia* or neuritis or neurodinya* or (nerve* **adj2** (pain* or injur* or disease* or affect* or damage* or degeneration*)) or causalgia* or hyperesthesi* or paresthesi* or hypesthesi* or paralysis or polyneuropath* or polyradiculo* or polyneuritis or burning feet syndrome or metatarsalgia).**ti,ab** | |
| **12** | neurologic disease/ | |
| **13** | ((neurologic* **adj2** (manifestation* or deficit* or sign or signs or symptom* or finding* or dysfunction*))).**ti,ab** | |
| **14** | sensory dysfunction/ OR exp somatosensory disorder/ OR perceptive threshold/ OR differential threshold/ OR pain threshold/ OR signal detection/ OR somatosensory evoked potential/ OR touch/ OR exp nociceptive pain/ OR referred pain/ OR exp nociception/ OR breakthrough pain/ OR face pain/ OR neurotoxicity/ OR hand foot syndrome/ | |
| **15** | (((sensation* or sense* or sensivity or sensorimotor or sensory or somatosensory or nociceptive or somatic or tissue* or tactile or touch* or pinprick or thermal or burning or shooting or deafferentation or differential or referred or percept* or facial or breakthrought) **adj3** (pain* or disorder* or disease* or disturb*or distortion* or discomfort* or dysfunction* or disruption* or impair* or symptom* or change* or abnormal* or mechanical or deficit* or limit* or loss or diminish* or reduc* or amplificat* or hyperesthetic or alter* or affect* or gating or | |

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| | | |
|---|---|---|
| | threshold*)) or pain threshold* or hyperalgesi* or allodynia* or formication* or tingling or hypersensitivity or oxyesthesi* or electric shock like or vibration* or nocicept* or neurotoxi* or hand foot syndrome*).**ti,ab** | |
| **16** | "biological phenomena and functions concerning organ systems"/ OR muscle disease/ OR neuromuscular disease/ OR muscle weakness/ OR muscle cramp/ OR arthralgia/ OR exp neurologic gait disorder/ OR proprioception/ OR weight bearing/ OR motor evoked potential/ | |
| **17** | ((motor* or muscle* or muscular or neuromuscular or functional* or movement*or walk* or gait* or balance or strength or tendon*or joint* or reflex* or tonus) **adj3** (pain* or disorder* or disturb* or discomfort* or dysfunction* or disruption* or impair* or symptom* or change* or abnormal* or mechanical or deficit* or limit* or loss or diminish* or reduc* or alter* or disabilit* or postural or weak* or fatigue) or propriocepti* or arthralgia* or myalgia* or polyarthralgia* or numbness or foot drop or footdrop or cramp*).**ti,ab**. | |
| **18** | **OR/10-17** | **2081962** |
| **CIPN** | | |
| **19** | chemotherapy-induced peripheral neuropathy/ | |
| **20** | ((chemotherapy induced **adj2** neuropath*) or CIPN).**ti,ab** | |
| **21** | **19 OR 20** | **2860** |
| **Antineoplastic agents** | | |
| **22** | exp antineoplastic agent/ OR combination chemotherapy/ | |
| **23** | (antineoplastic* or chemotherap* or ((anticancer or antitumo?) **adj3** (agent or agents or drug*))).**ti,ab** | |
| **24** | **22 OR 23** | **2717079** |
| **25** | **3 AND 9 AND 18 AND 24** | **5799** |
| **26** | **3 AND 9 AND 21** | **426** |
| **27** | **25 OR 26** | **5802** |
| **Exclusion Filters** | | |
| **28** | (editorial or letter).pt | |
| **29** | literature/ or case report/ or directory/ or editorial/ or interview/ or law/ or publication/ or consensus development/ or practice guideline/ | |
| **30** | **28 OR 29** | **5107933** |
| **31** | **27 NOT 30** | **5188** |

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**CINAHL:**

| Psychometrics properties | | Results |
|---|---|---|
| 1 | **MH**("Instrument Construction+" OR "Validation Studies" OR "Reproducibility of Results+" OR "Psychometrics+" OR "Observer Bias+" OR "Discriminant Analysis+") | |
| 2 | **TI**(reproducib* OR psychometr* OR clinimetr* OR clinometr* OR observer variation OR reliab* OR valid* OR coefficient OR internal consistency OR (cronbach* AND (alpha OR alphas)) OR item correlation* OR item selection* OR item reduction* OR agreement OR precision OR imprecision OR precise values OR test retest OR (test AND retest) OR (reliab* AND (test OR retest)) OR stability OR interrater OR inter rater OR intrarater OR intra rater OR intertester OR inter tester OR intratester OR intra tester OR interobserver OR inter observer OR intraobserver OR intra observer OR intertechnician OR inter technician OR intratechnician OR intra technician OR interexaminer OR inter examiner OR intraexaminer OR intra examiner OR interassay OR inter assay OR intraassay OR intra assay OR interindividual OR inter individual OR intraindividual OR intra individual OR interparticipant OR inter participant OR intraparticipant OR intra participant OR kappa OR kappa's OR kappas OR coefficient of variation OR repeatab* OR ((replicab* OR repeated) AND (measure OR measures OR findings OR result OR results OR test OR tests)) OR generaliza* OR generalisa* OR concordance OR (intraclass AND correlation*) OR discriminative OR known group OR factor analysis OR factor analyses OR factor structure OR factor structures OR dimensionality OR subscale* OR multitraitscaling analysis OR multitrait scaling analyses OR item discriminant OR interscale correlation* OR ((error OR errors) AND (measure*OR correlat* OR evaluat* OR accuracy OR accurate OR precision OR mean)) OR individual variability OR interval variability OR rate variability OR variability analysis OR (uncertainty AND (measurement OR measuring)) OR standard error of measurement OR sensitiv* OR responsive* OR (limit AND detection) OR minimal detectable concentration OR interpretab* OR (small* AND (real OR detectable) AND (change OR difference)) OR meaningful change OR minimal important change OR minimal important difference OR minimally important change OR minimally important difference OR minimal detectable change OR minimal detectable difference OR minimally detectable change OR minimally detectable difference OR minimal real change OR minimal real difference OR minimally real change OR minimally real difference OR ceiling effect OR floor effect OR Item response model OR IRT OR Rasch OR Differential item functioning OR DIF OR computer adaptive testing OR itembank OR cross cultural equivalence) | |
| 3 | **AB(**reproducib* OR psychometr* OR clinimetr* OR clinometr* OR observer variation OR reliab* OR valid* OR coefficient OR internal consistency OR (cronbach* AND (alpha OR alphas)) OR item correlation* OR item selection* OR item reduction* OR agreement OR precision OR imprecision OR precise values OR test retest OR (test AND retest) OR (reliab* AND (test OR retest)) OR stability OR interrater OR inter rater OR | |

| | | |
|---|---|---|
| | intrarater OR intra rater OR intertester OR inter tester OR intratester OR intra tester OR interobserver OR inter observer OR intraobserver OR intra observer OR intertechnician OR inter technician OR intratechnician OR intra technician OR interexaminer OR inter examiner OR intraexaminer OR intra examiner OR interassay OR inter assay OR intraassay OR intra assay OR interindividual OR inter individual OR intraindividual OR intra individual OR interparticipant OR inter participant OR intraparticipant OR intra participant OR kappa OR kappa's OR kappas OR coefficient of variation OR repeatab* OR ((replicab* OR repeated) AND (measure OR measures OR findings OR result OR results OR test OR tests)) OR generaliza* OR generalisa* OR concordance OR (intraclass AND correlation*) OR discriminative OR known group OR factor analysis OR factor analyses OR factor structure OR factor structures OR dimensionality OR subscale* OR multitraitscaling analysis OR multitrait scaling analyses OR item discriminant OR interscale correlation* OR ((error OR errors) AND (measure*OR correlat* OR evaluat* OR accuracy OR accurate OR precision OR mean)) OR individual variability OR interval variability OR rate variability OR variability analysis OR (uncertainty AND (measurement OR measuring)) OR standard error of measurement OR sensitiv* OR responsive* OR (limit AND detection) OR minimal detectable concentration OR interpretab* OR (small* AND (real OR detectable) AND (change OR difference)) OR meaningful change OR minimal important change OR minimal important difference OR minimally important change OR minimally important difference OR minimal detectable change OR minimal detectable difference OR minimally detectable change OR minimally detectable difference OR minimal real change OR minimal real difference OR minimally real change OR minimally real difference OR ceiling effect OR floor effect OR Item response model OR IRT OR Rasch OR Differential item functioning OR DIF OR computer adaptive testing OR itembank OR cross cultural equivalence) | |
| **4** | **1 OR 2 OR 3** | **1194447** |
| **Patient-Reported Outcome Measurements** | | |
| **5** | **MH**("Patient-Reported Outcomes+" OR "Surveys+" OR "Questionnaires+ OR "Self Report+" OR "Quality of Life+") | |
| **6** | **TI((**(patient# OR self) **N2** (report* OR relate* OR declar* OR rating* OR assess* OR examin* OR monitor* OR administer* OR rate* OR rating* or based or appraisal* or appraised or evaluat*) **N2** (outcome* OR symptom*)) or PROM* Or PRO or PROS OR (quality **N2** life) OR HRQL or HRQoL or QoL OR HR PRO or HRPRO) | |
| **7** | **AB**(((patient# OR self) **N2** (report* OR relate* OR declar* OR rating* OR assess* OR examin* OR monitor* OR administer* OR rate* OR rating* or based or appraisal* or appraised or evaluat*) **N2** (outcome* OR symptom*)) or PROM* Or PRO or PROS OR (quality **N2** life) OR HRQL or HRQoL or QoL OR HR PRO or HRPRO) | |

| | | |
|---|---|---|
| **8** | **TI**((disability or function* or subjective or utility or utilities or wellbeing or well being) **N3** (outcome or outcomes or index or indices or instrument or instruments or measure or measures or questionnaire or questionnaires or profile or profiles or scale or scales or score or scores or scoring or status or survey or surveys or measure*)) | |
| **9** | **AB**((disability or function* or subjective or utility or utilities or wellbeing or well being) **N3** (outcome or outcomes or index or indices or instrument or instruments or measure or measures or questionnaire or questionnaires or profile or profiles or scale or scales or score or scores or scoring or status or survey or surveys or measure*)) | |
| **Clinician reported outcomes measurements** | | |
| **10** | (**MH** "Outcome Assessment") | |
| **11** | **TI**(Clinician* reported outcome* or CROS or clinRO or clinROs OR ((clinical or clinician* or expert* or health professional* OR physician* OR doctor*) **N2** (judgement* or interpretation* or observation* or assessment* or measure*))) | |
| **12** | **AB**(Clinician* reported outcome* or CROS or clinRO or clinROs OR ((clinical or clinician* or expert* or health professional* OR physician* OR doctor*) **N2** (judgement* or interpretation* or observation* or assessment* or measure*))) | |
| **13** | **OR/5-12** | **924590** |
| **Peripheral neuropathy** | | |
| **14** | **MH**(peripheral nervous system diseases or brachial plexus neuropathies+ or complex regional pain syndrome or neuralgia or metatarsalgia or neuritis or polyneuropathies or polyradiculopathy or small fiber neuropathy or neural conduction or peripheral nerves+) | |
| **15** | **TI**(PNS disease* or peripheral nervous system disease* or neuropath* or complex regional pain syndrome* or neuralgia* or neuritis or neurodinya* or (nerve* **N2** (pain* or injur* or disease* or affect* or damage* or degeneration*)) or causalgia* or hyperesthesi* or paresthesi* or hypesthesi* or paralysis or polyneuropath* or polyradiculo* or polyneuritis or burning feet syndrome or metatarsalgia) | |
| **16** | **AB**(PNS disease* or peripheral nervous system disease* or neuropath* or complex regional pain syndrome* or neuralgia* or neuritis or neurodinya* or (nerve* **N2** (pain* or injur* or disease* or affect* or damage* or degeneration*)) or causalgia* or hyperesthesi* or paresthesi* or hypesthesi* or paralysis or polyneuropath* or polyradiculo* or polyneuritis or burning feet syndrome or metatarsalgia) | |
| **17** | **MH**(neurologic manifestations) | |
| **18** | TI((neurologic* **N2** (manifestation* or deficit* or sign or signs or symptom* or finding* or dysfunction*))) | |
| **19** | AB((neurologic* **N2** (manifestation* or deficit* or sign or signs or symptom* or finding* or dysfunction*))) | |

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| | | |
|---|---|---|
| **20** | MH (sensation disorders+ or sensation+ or evoked potentials+ or nociceptive pain+ or breakthrough pain or facial pain or referred pain or neurotoxicity syndromes or hand-foot syndrome) | |
| **21** | **TI**(((sensation* or sense* or sensivity or sensorimotor or sensory or somatosensory or nociceptive or somatic or tissue* or tactile or touch* or pinprick or thermal or burning or shooting or deafferentation or differential or referred or percept* or facial or breakthrought) **N3** (pain* or disorder* or disease* or disturb*or distortion* or discomfort* or dysfunction* or disruption* or impair* or symptom* or change* or abnormal* or mechanical or deficit* or limit* or loss or diminish* or reduc* or amplificat* or hyperesthetic or alter* or affect* or gating or threshold*)) or pain threshold* or hyperalgesi* or allodynia* or formication* or tingling or hypersensitivity or oxyesthesi* or electric shock like or vibration* or nocicept* or neurotoxi* or hand foot syndrome*) | |
| **22** | **AB**(((sensation* or sense* or sensivity or sensorimotor or sensory or somatosensory or nociceptive or somatic or tissue* or tactile or touch* or pinprick or thermal or burning or shooting or deafferentation or differential or referred or percept* or facial or breakthrought) **N3** (pain* or disorder* or disease* or disturb*or distortion* or discomfort* or dysfunction* or disruption* or impair* or symptom* or change* or abnormal* or mechanical or deficit* or limit* or loss or diminish* or reduc* or amplificat* or hyperesthetic or alter* or affect* or gating or threshold*)) or pain threshold* or hyperalgesi* or allodynia* or formication* or tingling or hypersensitivity or oxyesthesi* or electric shock like or vibration* or nocicept* or neurotoxi* or hand foot syndrome*) | |
| **23** | **MH**(musculoskeletal, neural, and ocular physiology or muscle fatigue or muscle strength+ or muscle tonus or muscular diseases or neuromuscular manifestations or reflex or muscle weakness or muscle cramp or arthralgia or gait disorders, neurologic+ or motor activity or physical mobility or muscle pain or neuromuscular control or weight-bearing or weight shifting) | |
| **24** | **TI**((motor* or muscle* or muscular or neuromuscular or functional* or movement*or walk* or gait* or balance or strength or tendon*or joint* or reflex* or tonus) **N3** (pain* or disorder* or disturb* or discomfort* or dysfunction* or disruption* or impair* or symptom* or change* or abnormal* or mechanical or deficit* or limit* or loss or diminish* or reduc* or alter* or disabilit* or postural or weak* or fatigue) or propriocepti* or arthralgia* or myalgia* or polyarthralgia* or numbness or foot drop or footdrop or cramp*) | |
| **25** | **AB**((motor* or muscle* or muscular or neuromuscular or functional* or movement*or walk* or gait* or balance or strength or tendon*or joint* or reflex* or tonus) **N3** (pain* or disorder* or disturb* or discomfort* or dysfunction* or disruption* or impair* or symptom* or change* or abnormal* or mechanical or deficit* or limit* or loss or diminish* or reduc* or alter* or disabilit* or postural or weak* or fatigue) or propriocepti* or arthralgia* or myalgia* or polyarthralgia* or numbness or foot drop or footdrop or cramp*) | |
| **26** | **OR/14-25** | **511082** |
| **CIPN** | | |

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|---|---|---|
| **27** | **TI**((chemotherapy induced N2 neuropath*) or CIPN) | |
| **28** | **AB**((chemotherapy induced N2 neuropath*) or CIPN) | |
| **29** | **27 OR 28** | **945** |
| **Antineoplastic agents** | | |
| **30** | **MH**(antineoplastic agents+) | |
| **31** | **TI**(antineoplastic* or chemotherap* or ((anticancer or antitumo?) N3 (agent or agents or drug*))) | |
| **32** | **AB**(antineoplastic* or chemotherap* or ((anticancer or antitumo?) N3 (agent or agents or drug*))) | |
| **33** | **30 OR 31 OR 32** | **178257** |
| **34** | **4 AND 13 AND 26 AND 33** | **1075** |
| **35** | **4 AND 13 AND 29** | **161** |
| **36** | **34 OR 35** | **1076** |
| **Exclusion Filters** | | |
| **37** | **PT**(biography OR case study OR commentary OR directories OR editorial OR interview OR legal case OR letter OR practice guidelines) | |
| **38** | **MH**("Biographies+" OR "Legislation" OR "News+" OR "Serial Publications+") | |
| **39** | **37 or 38** | **1405631** |
| **40** | **36 not 39** | **1050** |

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**PsycInfo (OVID)**

| | | |
|---|---|---|
| **Psychometrics properties** | | |
| **1** | exp psychometrics/ OR Interrater Reliability/ OR Discriminant Validity/ | **Results** |
| **2** | (reproducib* OR psychometr* OR clinimetr* OR clinometr* OR observer variation OR reliab* OR valid* OR coefficient OR internal consistency OR (cronbach* AND (alpha OR alphas)) OR item correlation* OR item selection* OR item reduction*).**ti,ab** OR (agreement OR precision OR imprecision OR precise values)**.tw** OR (test retest OR (test AND retest) OR (reliab* AND (test OR retest)) OR stability OR interrater OR inter rater OR intrarater OR intra rater OR intertester OR inter tester OR intratester OR intra tester OR interobserver OR inter observer OR intraobserver OR intra observer OR intertechnician OR inter technician OR intratechnician OR intra technician OR interexaminer OR inter examiner OR intraexaminer OR intra examiner OR interassay OR inter assay OR intraassay OR intra assay OR interindividual OR inter individual OR intraindividual OR intra individual OR interparticipant OR inter participant OR intraparticipant OR intra participant OR kappa OR kappa's OR kappas OR coefficient of variation).**ti,ab** OR (repeatab* OR ((replicab* OR repeated) AND (measure OR measures OR findings OR result OR results OR test OR tests))).**tw** OR (generaliza* OR generalisa* OR concordance OR (intraclass AND correlation*) OR discriminative OR known group OR factor analysis OR factor analyses OR factor structure OR factor structures OR dimensionality OR subscale* OR multitraitscaling analysis OR multitrait scaling analyses OR item discriminant OR interscale correlation* OR ((error OR errors) AND (measure*OR correlat* OR evaluat* OR accuracy OR accurate OR precision OR mean)) OR individual variability OR interval variability OR rate variability OR variability analysis OR (uncertainty AND (measurement OR measuring)) OR standard error of measurement OR sensitiv* OR responsive* OR (limit AND detection) OR minimal detectable concentration OR interpretab* OR (small* AND (real OR detectable) AND (change OR difference)) OR meaningful change OR minimal important change OR minimal important difference OR minimally important change OR minimally important difference OR minimal detectable change OR minimal detectable difference OR minimally detectable change OR minimally detectable difference OR minimal real change OR minimal real difference OR minimally real change OR minimally real difference OR ceiling effect OR floor effect OR Item response model OR IRT OR Rasch OR Differential item functioning OR DIF OR computer adaptive testing OR itembank OR cross cultural equivalence).**ti,ab** | |
| **3** | **1 OR 2** | **897151** |
| **Patient-Reported Outcome Measurement instruments (PROM)** | | |
| **4** | "Patient Reported Outcome Measures"/ or Surveys/ OR Questionnaires/ OR Self-Report/ OR exp Quality of Life/ OR Quality of Life Measures/ | |

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| | | |
|---|---|---|
| **5** | ((patient? OR self) adj2 (report* OR relate* OR declar* OR rating* OR assess* OR examin* OR monitor* OR administer* OR rate* OR rating* or based or appraisal* or appraised or evaluat*) adj2 (outcome* OR symptom*) or PROM? Or PRO or PROS OR (quality adj2 life) OR HRQL or HRQoL or QL or QoL OR HR-PRO or HRPRO).**ti,ab** | |
| **6** | ((disability or function* or subjective or utility or utilities or wellbeing or well being) adj3 (outcome or outcomes or index or indices or instrument or instruments or measure or measures or questionnaire or questionnaires or profile or profiles or scale or scales or score or scores or scoring or status or survey or surveys or measure*)).**ti,ab** | |
| **Clinician reported outcomes measur** | | |
| **7** | Health Outcomes/ | |
| **8** | (Clinician* reported outcome* or CROS or clinRO or clinROs OR ((clinical or clinician* or expert* or health professional* OR physician* OR doctor*) adj2 (judgement* or interpretation* or observation* or assessment* or measure*))).**ti,ab** | |
| **9** | **OR/4-8** | **261776** |
| **Peripheral neuropathy** | | |
| **10** | Peripheral Neuropathy/ OR Neuropathic Pain/ OR Neuralgia/ OR "Complex Regional Pain Syndrome (Type I)"/ OR Central Nervous System Disorders/ | |
| **11** | (PNS disease* or peripheral nervous system disease* or neuropath* or complex regional pain syndrome* or neuralgia* or neuritis or neurodinya* or (nerve* adj2 (pain* or injur* or disease* or affect* or damage* or degeneration*)) or causalgia* or hyperesthesi* or paresthesi* or hypesthesi* or paralysis or polyneuropath* or polyradiculo* or polyneuritis or burning feet syndrome or metatarsalgia).**ti,ab** | |
| **12** | Nervous System Disorders/ | |
| **13** | ((neurologic* adj2 (manifestation* or deficit* or sign or signs or symptom* or finding* or dysfunction*))).**ti,ab** | |
| **14** | Somatosensory Disorders/ OR Pressure Sensation/ OR Pain Perception/ OR "Sensory Disabilities (Attitudes Toward)"/ OR Sensory Integration Dysfunction/ OR Sensory Gating/ OR Perceptual Disturbances/ OR Sensory System Disorders/ OR Pain Thresholds/ OR Thresholds/ OR Somatosensory Evoked Potentials/ OR "Signal Detection (Perception)"/ OR Evoked Potentials/ OR Somesthetic Perception/ OR Tactual Perception/ OR Somatoform Pain Disorder/ OR Somatoform Disorders/ OR Neurotoxicity/ | |
| **15** | (((sensation* or sense* or sensivity or sensorimotor or sensory or somatosensory or nociceptive or somatic or tissue* or tactile or touch* or pinprick or thermal or burning or shooting or deafferentation or differential or referred or percept* or facial or breakthrought) adj3 (pain* or disorder* or disease* or disturb*or distortion* or discomfort* or dysfunction* or disruption* or impair* or symptom* or change* or abnormal* or mechanical or deficit* or limit* or loss or diminish* or reduc* or amplificat* or hyperesthetic or alter* or affect* or gating or | |

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| | threshold*)) or pain threshold* or hyperalgesi* or allodynia* or formication* or tingling or hypersensitivity or oxyesthesi* or electric shock like or vibration* or nocicept* or neurotoxi* or hand foot syndrome*).**ti,ab** | |
| **16** | Musculoskeletal Disorders/ OR Neuromuscular Disorders/ OR Muscular Disorders/ OR Myasthenia/ OR Articulation Disorders/ OR Movement Disorders/ OR Proprioception/ | |
| **17** | ((motor* or muscle* or muscular or neuromuscular or functional* or movement*or walk* or gait* or balance or strength or tendon*or joint* or reflex* or tonus) adj3 (pain* or disorder* or disturb* or discomfort* or dysfunction* or disruption* or impair* or symptom* or change* or abnormal* or mechanical or deficit* or limit* or loss or diminish* or reduc* or alter* or disabilit* or postural or weak* or fatigue) or propriocepti* or arthralgia* or myalgia* or polyarthralgia* or numbness or foot drop or footdrop or cramp*).**ti,ab**. | |
| **18** | **OR/10-17** | **298714** |
| **CIPN** | | |
| **19** | ((chemotherapy induced adj2 neuropath*) or CIPN).**ti,ab** | |
| **Antineoplastic agents** | | |
| **20** | Antineoplastic Drugs/ OR Chemotherapy/ | |
| **21** | (antineoplastic* or chemotherap* or ((anticancer or antitumo?) adj3 (agent or agents or drug*))).**ti,ab** | |
| **22** | **20 OR 21** | **7387** |
| **23** | **3 AND 9 AND 18 AND 22** | **94** |
| **24** | **3 AND 9 AND 19** | **28** |
| **25** | **23 OR 24** | **94** |
| **Exclusion Filters** | | |
| **26** | Exp Biography/ OR Case Report/ OR exp laws/ OR Interviews/ OR Newspapers/ OR Treatment Guidelines/ OR Popular Culture/ | |
| **27** | **25 not 26** | **93** |

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